

- Traffic speed is kept low to minimise damage to the work, i.e. a newly laid seal.

7.6.6 Workers on foot

When work activities require workers on foot, the minimum clearances outlined in Section 4.3.4 Minimum clearances of workers to traffic must be maintained at all times. In addition, in accordance with Section 4.5 Speed zones, a 30 km/h speed zone should be installed where workers are closer than 1.5 m to moving traffic.

Additionally, to give maximum protection to workers on foot where temporary safety barriers cannot be provided:

- Work area delineation should be provided where possible;
- Work vehicles should be placed to shield workers from the path of errant vehicles; and
- Work vehicles and shadow vehicles should be positioned in line with the boundary of work area which is more exposed to traffic;

When attending parked work vehicles, the vehicles may be accessed from the side or the rear if there is no risk of errant vehicles impacting on the rear of the vehicle further away from traffic.

7.7 Designing a TGS: Static work

7.7.1 General

When designing a TGS for static works, the size and position of the work area must enable the work site to be managed effectively for the safety of road workers and road users. The size and position might not be consistent for the whole duration of work depending on changes in project tasks or location. When designing a TGS for static works:

- Temporary speed limits must be in accordance with Section 4.5 Speed zones and be limited to the immediate times and locations required;
- Sign placement must be in accordance with Section 6.5.5 Sign placement with signs that are not immediately relevant, removed or covered;
- Lane widths must be in accordance with Section 4.3.2 Lane widths; and

Protection of the work area must be in accordance with Section 4.3.5 Protection of work area.

When designing a TGS for a static work site, the work area must be designed to provide for:

- Access and egress of workers, plant and equipment;
- Storage areas for materials and equipment;
- Safe sight distances for signage, traffic control devices and the work area; and
- Fixed work areas, which must be marked by barrier boards or delineation devices where applicable.

For a long term static work site, a number of TGS may be needed in addition to the use of staging plans.

7.7.2 Around TTM methods

7.7.2.1 General

Around traffic management strategies include detours, road closures or the use of crossovers protected by a separated median. Around TTM methods provide the highest level of worker protection as the risk of live traffic is removed. Around TTM strategies should be considered wherever possible. Prior to the

implementation of any of these strategies, approvals must have been gained with appropriate road authority and emergency service consultation and notification prior to installing the treatment.

7.7.2.2 Road closures

A road closure involves the complete closure of all trafficable lanes to all road users to ensure the safety of road workers and road users whilst works are underway. For a full road closure to be undertaken, it must have been identified and approved as part of the TMP. When designing a TGS for a road closure:

- The ROAD CLOSED (T2-4) sign must be used in conjunction with barrier boards at the start of the site to prohibit general road user access;
- Speed zones must be reduced approaching the closure;
- The number of traffic lanes should be progressively reduced on approach to the closure. For example, on a two-lane one-way road, on approach to the closure, the left lane should be closed before the right lane;
- If an approach or exit lane is closed at an intersection, the corresponding approach or exit lane on the opposite side of the intersection must also be closed;
- Traffic control vehicles may be used to physically obstruct access to closed roads to supplement delineation devices if delineation devices alone are insufficient;
- If a motorway exit is temporarily closed, closure signs must be installed in advance of:
 - The previous exit if traffic is detoured through said previous exit; or
 - The closed exit if traffic is detoured through the next exit.
- Closures at on and off-ramps must be reinforced with a line of traffic cones at appropriate site distance from the work area. The traffic cones should be placed in a continuous line or chicaned to slow road users, assist in re-enforcing the closure and define work vehicle entry points. Traffic cones must be placed 1 m apart to stop road users from driving through, and advance warning signs are provided as detailed above.

Where access to properties is affected by the road closure, the following treatments should be considered:

- Provision of an alternative access point for residents and commercial businesses;
- Provision for local traffic to travel through the work area with a delineated path or guidance from a traffic controller. A LOCAL TRAFFIC ONLY sign should be erected beside the access point as applicable. Large gaps should not be left between delineation devices (e.g. traffic cones) at access points to avoid general traffic passing through by mistake or intentionally; or
- Allow pedestrian access only. Local residents will need to park elsewhere during this time.

Where road closures are utilised with detours required, detours should be installed in accordance with [Section 7.7.2.3 Detours](#).

[Figure 7-7](#) provides an example layout of a detour and full road closure scenario.

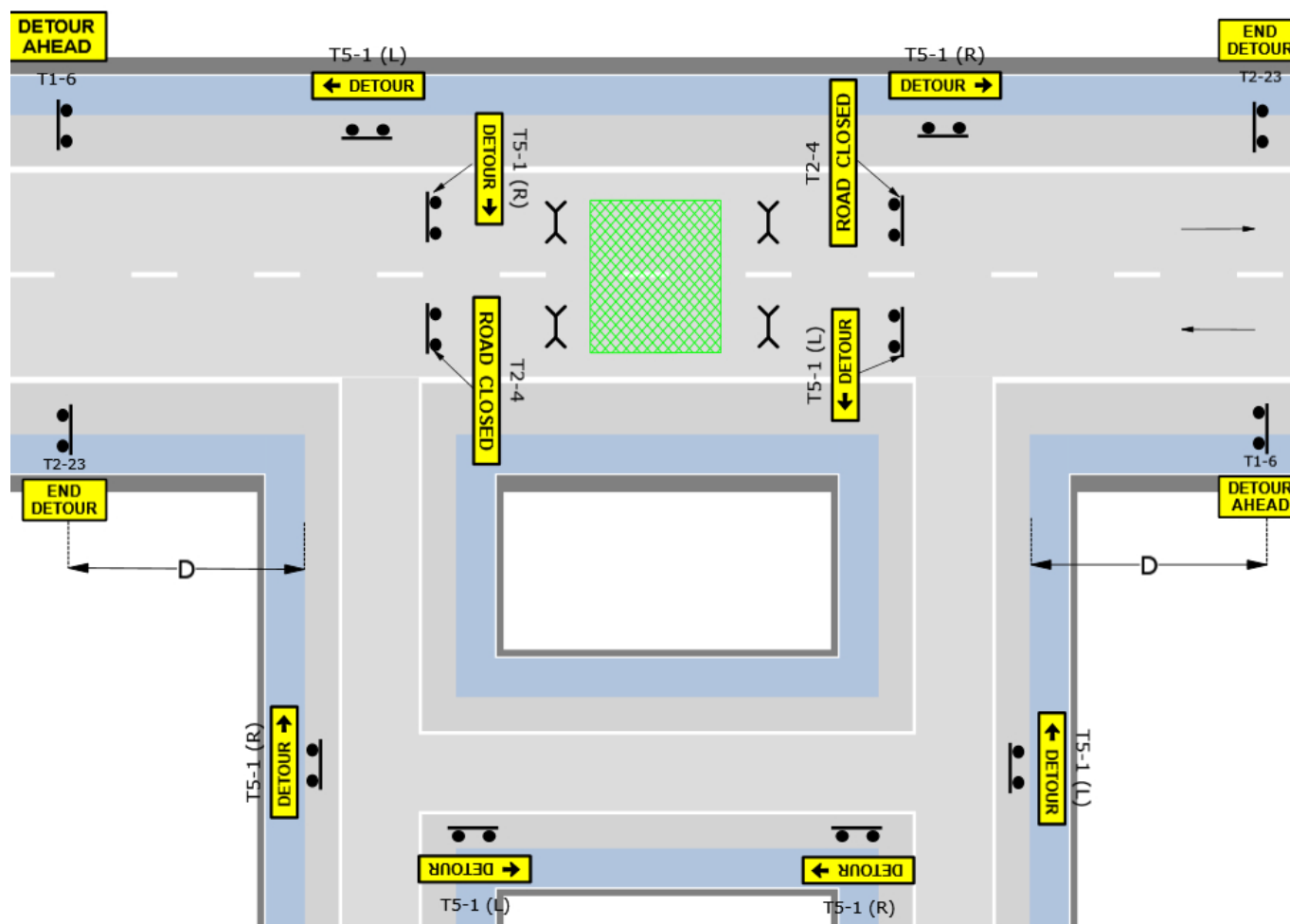


Figure 7-7. Example layout – detour

7.7.2.3 Detours

When designing a TGS for a static work site that utilises an around method, sufficient advanced warning signs and information to advise road users of the changed conditions must be provided. This includes the following:

- A Worker symbolic sign must be placed on the left side of the roadway in advance of the work area if road workers are visible to traffic; and
- A DETOUR AHEAD sign (or VMS) must be placed to give advance warning of an approaching route to detour around a work area. The first detour marker must be placed no more than 100 m past the DETOUR AHEAD sign (or VMS). Detour markers are mounted horizontally with the arrow either vertically upward, at 45 degrees upwards to the left or right, or horizontally to the left or right.

The advance warning area is critical to the success of an *around* traffic management strategy to reduce the risk of confusion to road users. Repeated information may assist road users to navigate their way safely and effectively to their intended destinations.

7.7.2.4 Side tracks and crossovers

When designing a TGS that includes a sidetrack or a crossover the following provisions apply (see [Figure 7-8](#) for an example layout with side track):

- Approval must be sought from the maintaining authority to use any side track, cross over or detour;

- The pavement must be of sufficient structural strength to carry the loads and traffic volumes expected;
- The pavement on detours must be monitored to ensure that any damage is quickly rectified;
- The width of a detour and its alignment must be adequate for the expected traffic types and volumes;
- Provision must be made for vulnerable road users and high risk pedestrians and public transport;
- Access must be provided for local traffic;
- Delineation must be provided on the alternate route as appropriate, and may include but is not limited to:
 - Linemarking;
 - Temporary guide posts;
 - Retro-reflective pavement markers (RRPMs), and
 - Signposting.
- Signposting must be clear and adequate; and
- DETOUR AHEAD (T1-6) signs must be used to give advance warning of any detours.

Additionally, the establishment of long-term sidetracks must undergo a detailed design and review process.

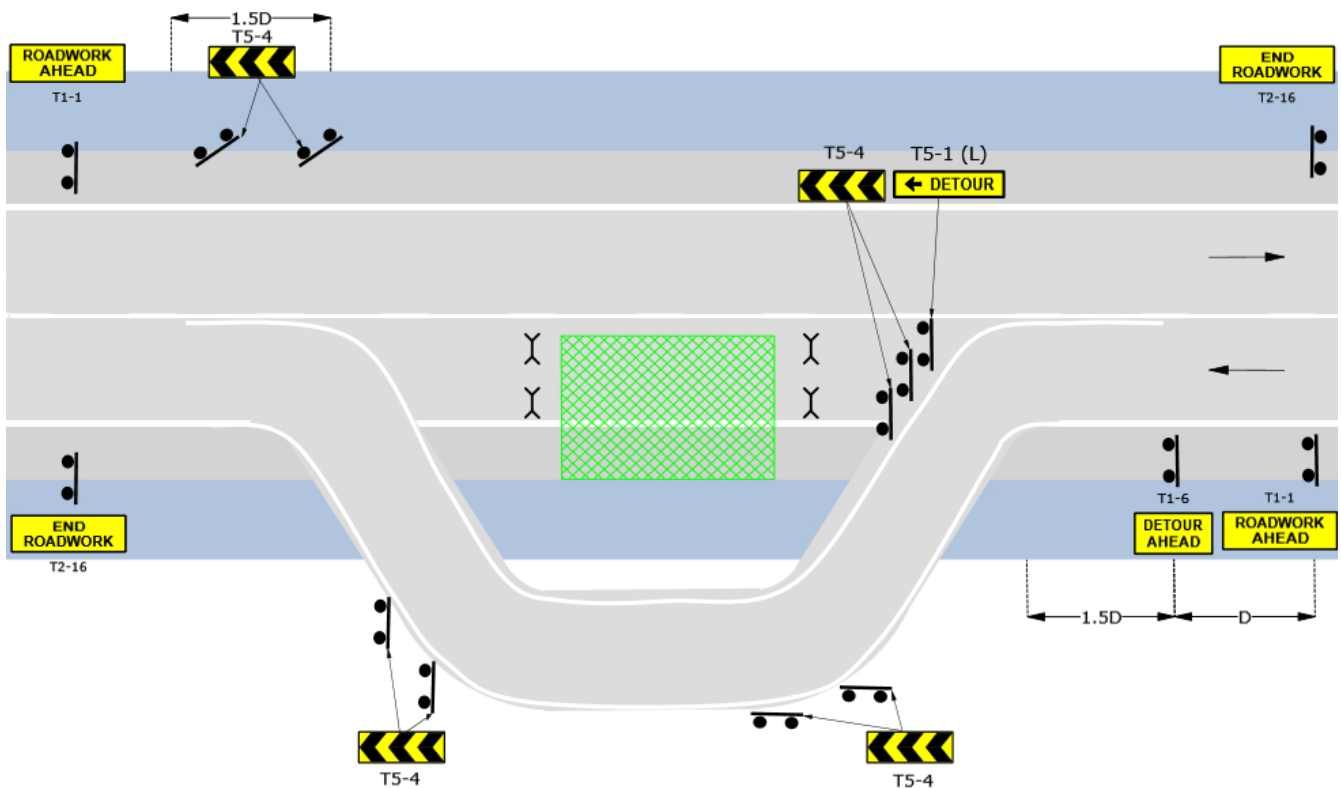


Figure 7-8. Example layout – side track

7.7.3 Past TTM methods

7.7.3.1 General

Where it is not possible to move traffic around a work area, the following forms of traffic past methods must be considered:

- Lane closure and lateral shift; or
- Protection of work site via accepted safety barriers as per Section 6.7 Temporary safety barriers.

7.7.3.2 Closing a lane

Procedure

When closing a traffic lane, signs and devices should be installed and removed in accordance with [Section 6.4 Sequence for installation and removal of signs and devices](#), see also [Figure 7-9](#). When closing off a lane, work must progress in the direction of traffic flow, starting at the farthest point from the job. The vehicle housing the traffic cones, with warning lights and hazard lights operating, must commence at the start of the taper and progressively place the layout in the direction of traffic flow until the taper is completed. The vehicle, and any workers on foot, must be protected by a shadow vehicle during installation of the lane closure.

Minimum requirements for all roads

A minimum of two temporary hazard markers (T5-4 or T5-5) per closed lane must be installed. Cones or bollards must also be used to create a taper to close the lane and define the travel path if traffic is to be merged into an adjacent lane, shifted laterally or be directed by a PTCD or a traffic controller past the work area. When working in low light or restricted visibility conditions (e.g. at night), consideration must be given to using a flashing arrow sign in addition to, or in place of, the temporary hazard markers.

If temporary delineation is needed on both sides of the vehicle path at a taper, hazard markers must only be used on the side that is primarily guiding traffic away from the most significant obstruction or other hazard. Hazard markers on both sides of a traffic path must not be placed as this can cause confusing patterns of delineation. Cones should be placed on the lane line between the end of the taper and the work area and on the other side of the travel path in accordance with [Section 6.2.5 Spacing of cones and bollards](#) and [Section 7.6.2.2 Tapers](#).

If the work area is beyond a crest or curve with limited visibility to the approaching motorist (less than 100 m) then the layout should be extended in advance of the crest or curve.

Additional requirements for multi lane roads

On multi-lane roads with existing posted speed limit of less than 85 km/h and less than 20,000 vpd consideration must be given to using a flashing arrow sign in addition to, or in place of, the temporary hazard markers. The TGS must also include:

- Appropriate lane status (T2-6-1 or T2-6-2) signs; and
- Where required by [Section 6.10 Truck and trailer-mounted attenuators \(TMA\)](#), the location of the TMA as part of the lane closure.

Additional requirements for motorways or other multi lane roads

On the motorways listed in [Table 6-20](#) or other multi lane roads with an existing posted speed limit of greater than 85 km/h and traffic volumes greater than 20,000 vpd, a flashing arrow sign must be provided at each merge taper in addition to the two (minimum) temporary hazard markers (T5-4 or T5-5).

The flashing arrow sign and markers must remain in position and be clearly visible from the commencement of the taper installation, at all times during the operation of the TGS and until the taper is removed.

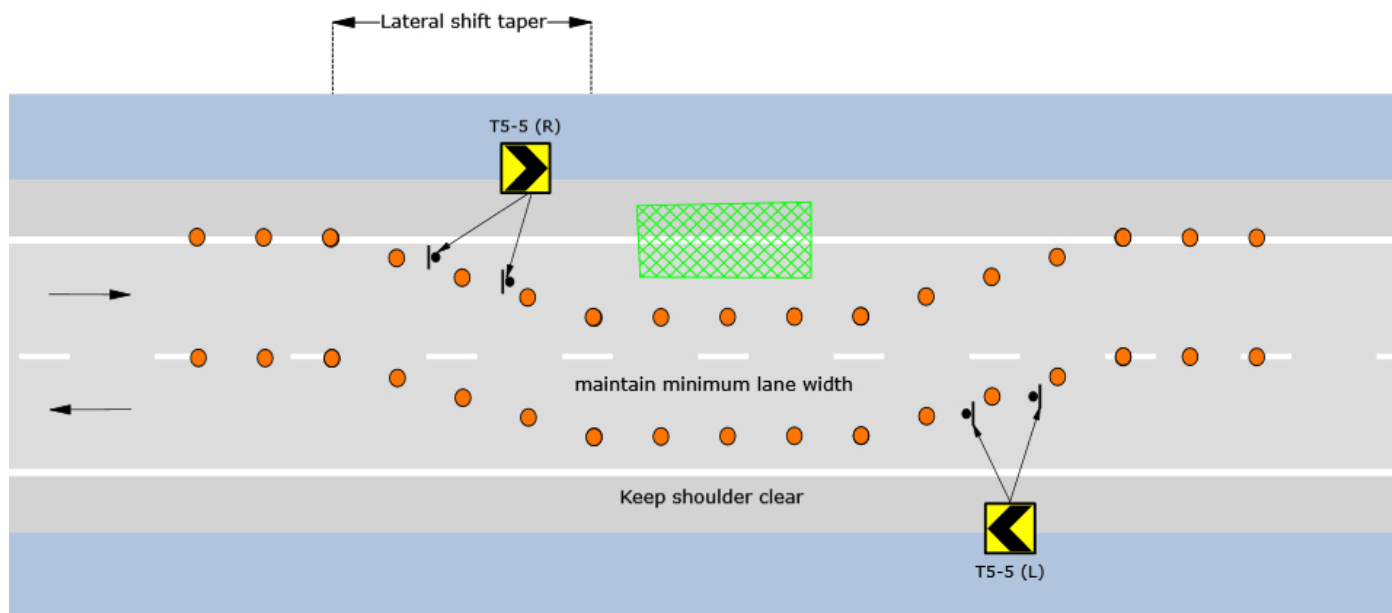


Figure 7-9. Example layout – static: lane closure, two-lane, two-way (formerly TCP 41)

Additional examples of lane closures are provided in [Appendix D – Work type layout examples](#).

7.7.3.3 Closing a shoulder

When closing a shoulder, a temporary hazard marker shall be placed 20 m before the work area with at least one additional marker, at 20 m spacing, in advance. The hazard markers are to be no less than 1 m from the edge line. A barrier board should be placed at the beginning and end of the work area. In addition a SHOULDER CLOSED (T2-19n) must be placed D in advance of the first temporary hazard marker (T5-4 or T5-5), (see also [Figure 7-10](#)).

Consideration must be given to cyclists, buses, resident's vehicle, parking and any other road user that may utilise the shoulder in this area, see [Section 4.4 Providing for specific road users](#).

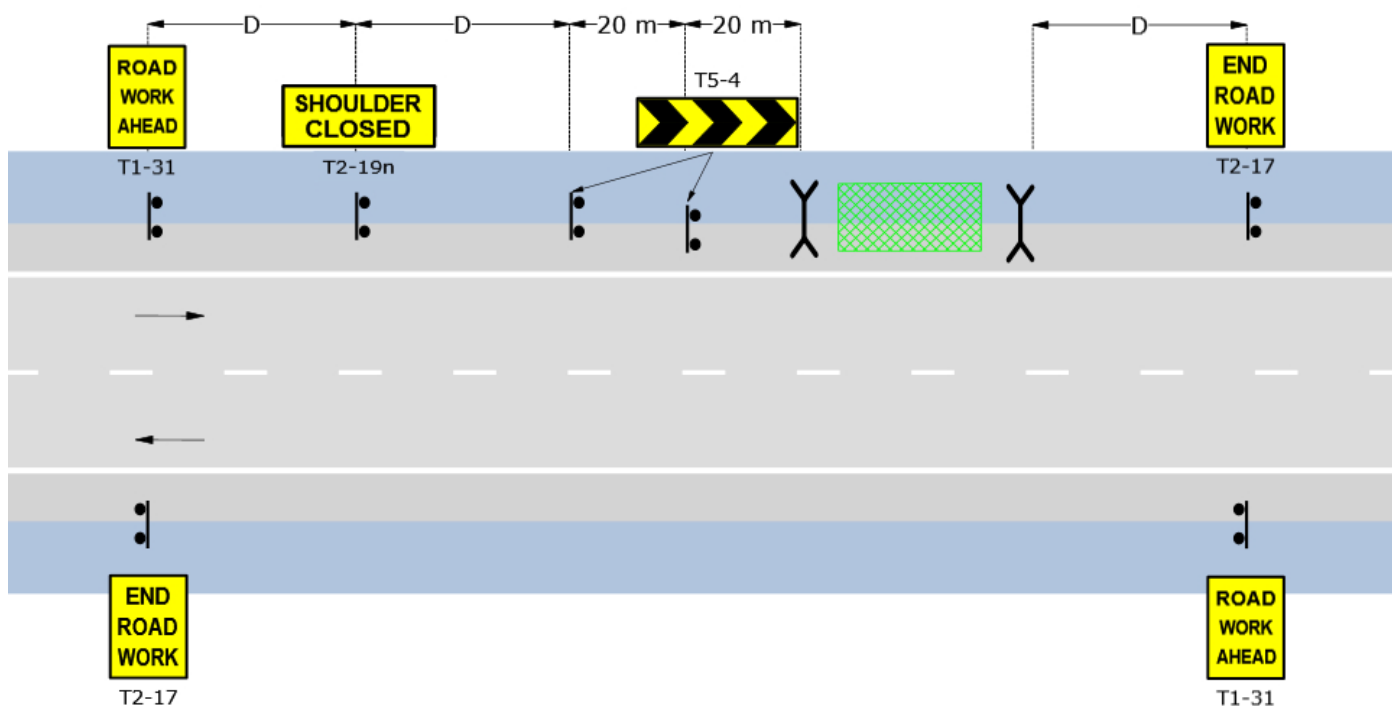


Figure 7-10. Example layout – shoulder closure

7.7.3.4 Single lane operation

Single lane operation may be used on two-way two-lane roads with a very low traffic volume and appropriate sight distance. GIVE WAY (R1-2) and ONE LANE (R9-9n) sign assembly must be used to assign priority to one direction of travel past the work area where the travelled way is reduced to less than that required for two lanes of traffic and no traffic controller is provided (see [Figure 7-11](#)). This is appropriate for work when:

- The traffic volume is 100 vpd or less and the traffic speed is less than 75 km/h;
- Each entry to the work area is visible from the other;
- The work area is less than 100 m long; or
- There is sight distance to opposing traffic of at least 300 m beyond the far end of the work area for traffic facing the GIVE WAY/ONE LANE assembly sign.

NO OVERTAKING OR PASSING (R6-1) sign must be erected at the start of the single lane for traffic in the opposite direction. See also [Section 6.5.9 Requirements for specific signs](#).

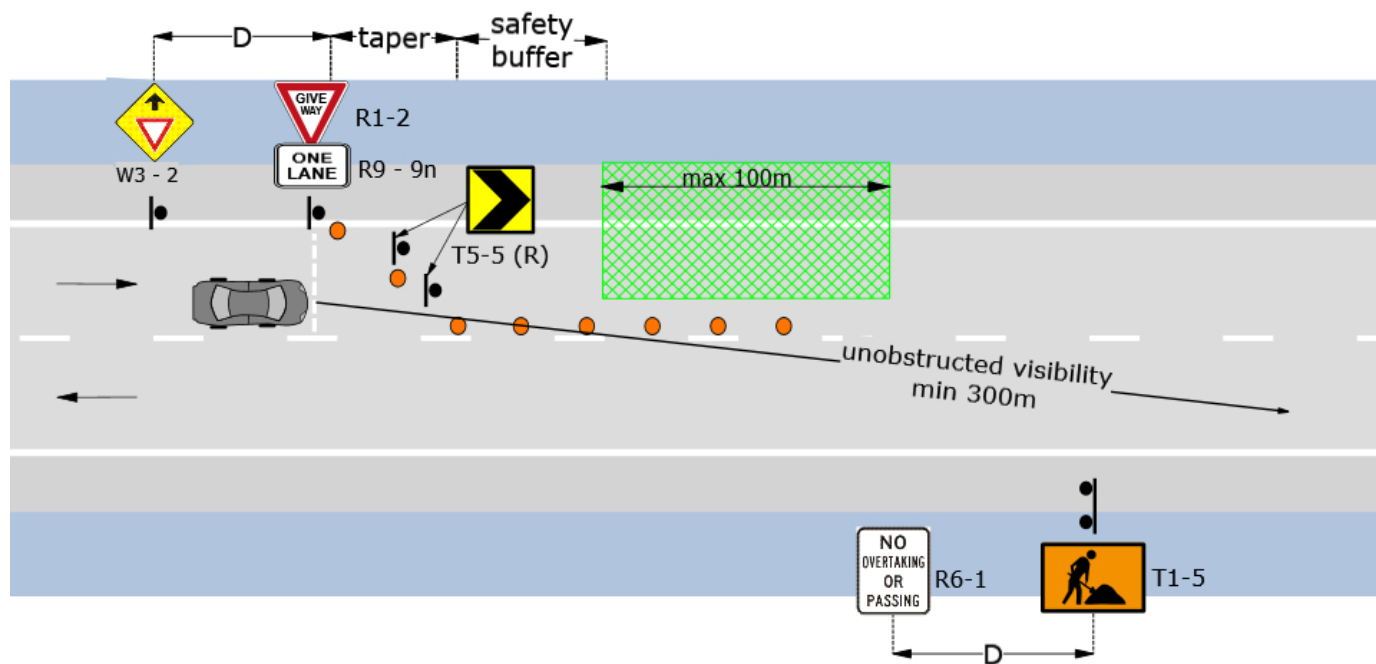


Figure 7-11. Example layout – single lane operation

7.7.3.5 Additional requirements for dual carriageway and multi-lane roads

When working on a central lane or in the median between carriageways there must be adequate lateral separation between workers on foot or plant from the opposing traffic flow. The lateral separation requirements in [Section 4.3.4 Minimum clearances of workers to traffic](#) should be applied at a minimum and in some situations an increased lateral separation may be required due to site-specific risks. The required lateral separation can be achieved by variety of means such as detouring the opposing traffic flow, providing a temporary barrier with a containment fence (for workers) or for shorter duration works by closing the opposing traffic lane.

Refer to [Section 7.7.3.2 Closing a lane](#) for the requirements for closing a lane on dual carriageway and multi-lane roads.

Where work is undertaken on a multi-lane dual carriageway:

- Signs and devices, excluding pavement arrows, must be placed on both sides of the carriageway, where practical, in accordance with [Section 6.5.6 Duplication of signs](#).
- If the carriageway has three or more lanes and two lanes are to be closed, then this should be done one lane at a time, with the distance between tapers at least $1.5D$, see [Figure 7-12](#); and
- Site access is established in accordance with [Section 5.2.3.2 Designing site access on multi-lane roads or high speed environments](#), where required.

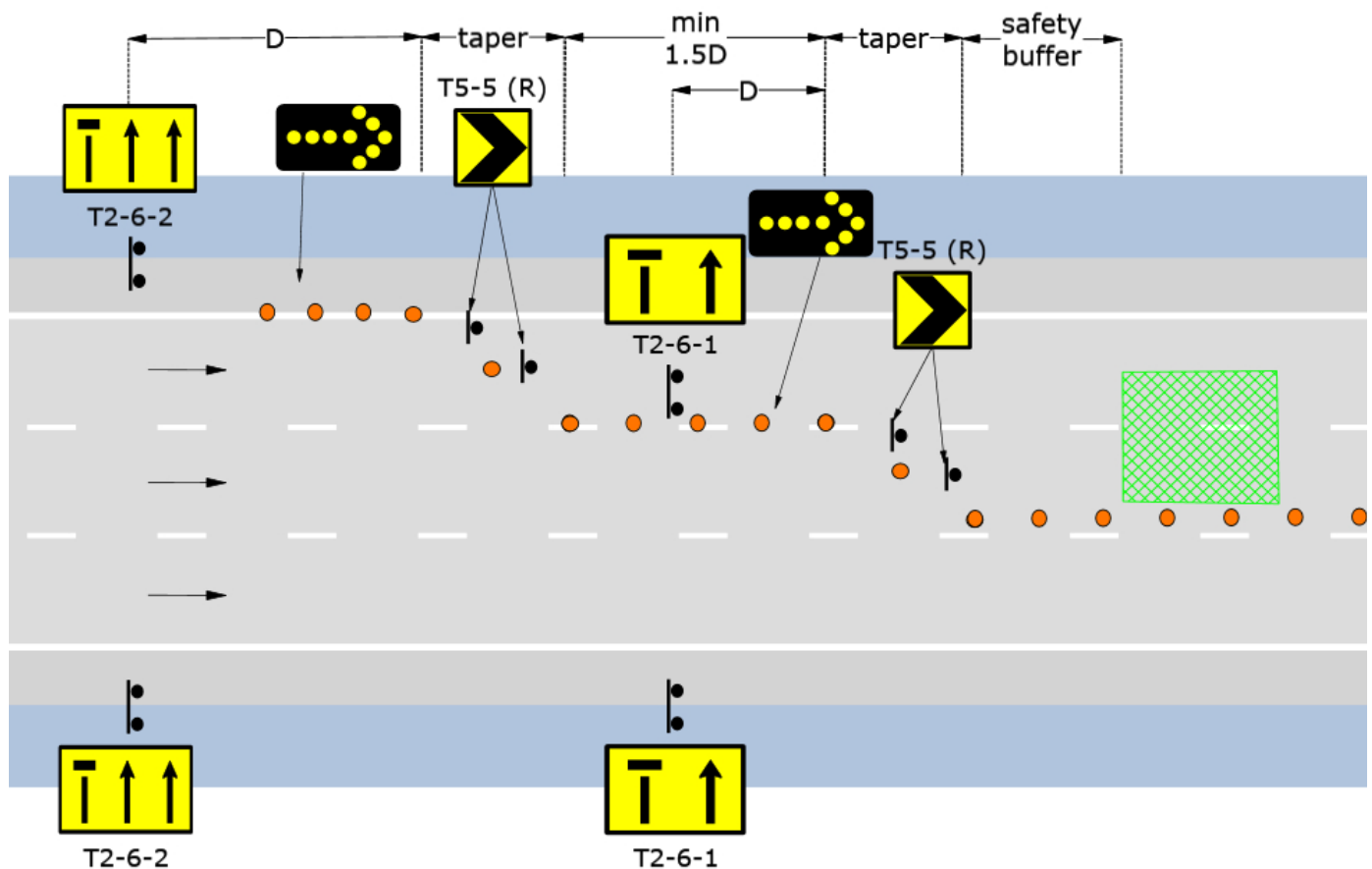


Figure 7-12. Example layout – static: lane closure, 3 lane

7.7.4 Through TTM methods

7.7.4.1 General

All static work sites that cannot have around or past controls applied must seek to ensure the highest order of available treatments are designed into the TGS, so as to reduce the potential for vehicles entering the work area erroneously. This should include speed reductions and delineation installed in accordance with [Section 4.3.5 Protection of work area](#) to reduce the risk of incidents between road workers and traffic.

7.7.4.2 Working at traffic signals

For short duration low impact works such as emergency fault repairs or routine maintenance, traffic controls need to be quickly set up and dismantled while still being effective in directing traffic past or through work areas. These types of works may be best managed by developing a suite of generic TGS that can be selected from, customised and implemented at various locations as site suitable TGS.

For the longer term or more major works such as pavement resurfacing, signal reconstructions or the construction of new traffic signals, site-specific TGS should be developed for each location and in some cases for each stage of work at the intersection.

In addition to the planning requirements in [Section 5.3.3 Working at or close to traffic signals](#) the following applies to designing a TGS for work at these locations:

- Any traffic control arrangements that will affect the operation of traffic signals or require them to be flashed yellow must be approved as part of the road occupancy licence. The Transport Management Centre (TMC) implements these approved changes to signal operation upon request during the implementation of the TGS;

- If working at or in close proximity to signals and alternate traffic control is needed, the signals must be flashed yellow and traffic control implemented on all approaches of the intersection prior to the use of a PTCB or STOP/SLOW bat;
- If turn restrictions or detours are necessary, in addition to the requirements of [Section 7.7.2 Around TTM methods](#) the following must be considered:
 - The impact that additional traffic has on the operation (phasing) of any traffic signals on the alternative route; and
 - The impact that any lane closures, particularly turn lanes has on signal operation (phasing) and intersection efficiency.
- Contact the TMC if real-time adjustments are needed to the operation (phasing) of traffic signals or adjacent signals during roadworks;
- Inform road users in advance of proposed works by advertising, for example using VMS roadside noticeboard advertising, letter drops or press releases; and
- Ensure that adequate provision is made for pedestrians and mobility aid users if it is necessary to turn off pedestrian signals or close a pedestrian crossing at intersection traffic signals.

Where lane closures are needed, the requirements of [Section 7.7.3.2 Closing a lane](#) must be applied.

[Figure 7-13](#) provides an example layout of working at traffic signals with lane closure.

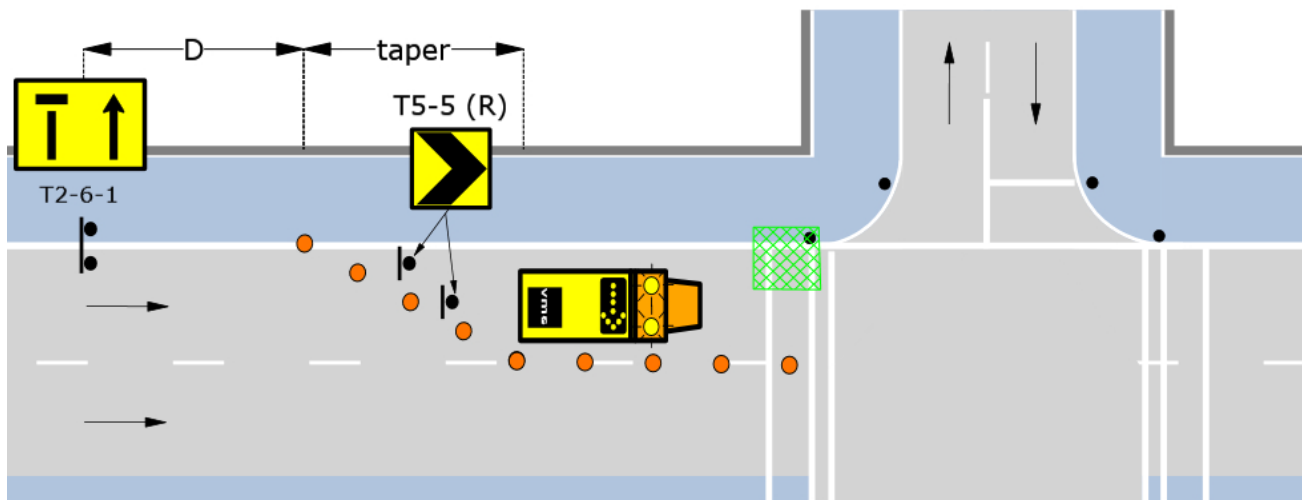


Figure 7-13. Example layout – work at traffic signals with lane closure

7.7.4.3 Excavations

Special consideration for the safety of both workers and traffic must be given while traffic flow is adjacent to excavations.

A temporary safety barrier must be installed where traffic travels beside an open excavation, deeper than 0.5 m and within 3 m of the travel path or edge line. When traffic is greater than 3 m from the excavation, the requirement for a temporary safety barrier should be considered during development of the TMP and risk assessment.

A temporary safety barrier must also be installed where the excavation depth exceeds 200 mm, and when the following applies:

- The excavation remains open longer than two weeks; and
- The distance of the excavation to the travel path or edge line is less than any of the following:
 - 3 m for 60 km/h approach speed;

- 6 m for 80 km/h approach speed; or
- 9 m for 100 km/h or greater approach speed.

Temporary safety barriers installed must be in addition to barriers for excavation fall protection of road workers and road users.

When the adjacent lane is not under traffic control, excavations shallower than 0.5 m and within 3 m of the travel path or edge line must be defined by:

- Plastic mesh fencing;
- Barrier boards perpendicular to the traffic flow; or
- Cones, bollards or similar delineation.

Additional controls or mitigation measures as identified in the TMP or risk assessment must also be installed, e.g. temporary safety barriers.

If the project manager considers that compliance with these requirements is not practical or warranted, a more thorough risk analysis must be made, based on Austroads *Guide to Road Design Part 6: Roadside Design, Safety and Barriers* and Transport *Supplement to AGRD, Part 6*, and an application for a departure be submitted in accordance with Section 2.8 Departures from this Technical Manual.

In planning and carrying out works, priority must be given to backfilling all excavations near traffic as soon as practicable.

7.7.4.4 Bituminous works

Bituminous works or other pavement surfacing works may be conducted under active traffic conditions or when the work area has been cleared of traffic. In most cases, there will be some interruption to traffic either from the work or from plant and vehicles associated with the work. This interruption to traffic should be kept to a minimum by careful planning and execution of the work. Such planning and control of the work also assists in the safe completion of the work.

The local community should be notified in advance of planned bitumen sealing works especially on busy roads.

The project manager must ensure that risks in the bituminous works are taken into account in planning the work so that traffic delay does not exceed a maximum of 15 minutes. This suggests an operations cycle of no longer than 10 minutes. In such cases, the following should be determined:

- Widths and lengths of area to be treated;
- Timing to avoid peak traffic periods;
- Whether to introduce side tracks or detours; and
- Temporary closure of intersecting roads.

In the case of bituminous spraying works, care should be taken to ensure that wet bitumen or loose stones do not cause a hazard to traffic. For this reason, the works should be monitored for some days and adjustments made to signs or temporary speed zones to suit the road conditions.

For signs at sprayed sealing works, the four stages that should be considered are:

- Before sealing begins;
- During the sealing operations;
- Until the last loose stones are removed; and
- Until the appropriate linemarking and road markings are fully restored.

See Figure 7-14 for signage examples.

Where sealing works are undertaken in areas where the existing permanent speed limit is greater than 55 km/h, a 60 km/h roadwork speed zone must be installed and remain in place until the number of loose aggregate particles falls to the specified level.

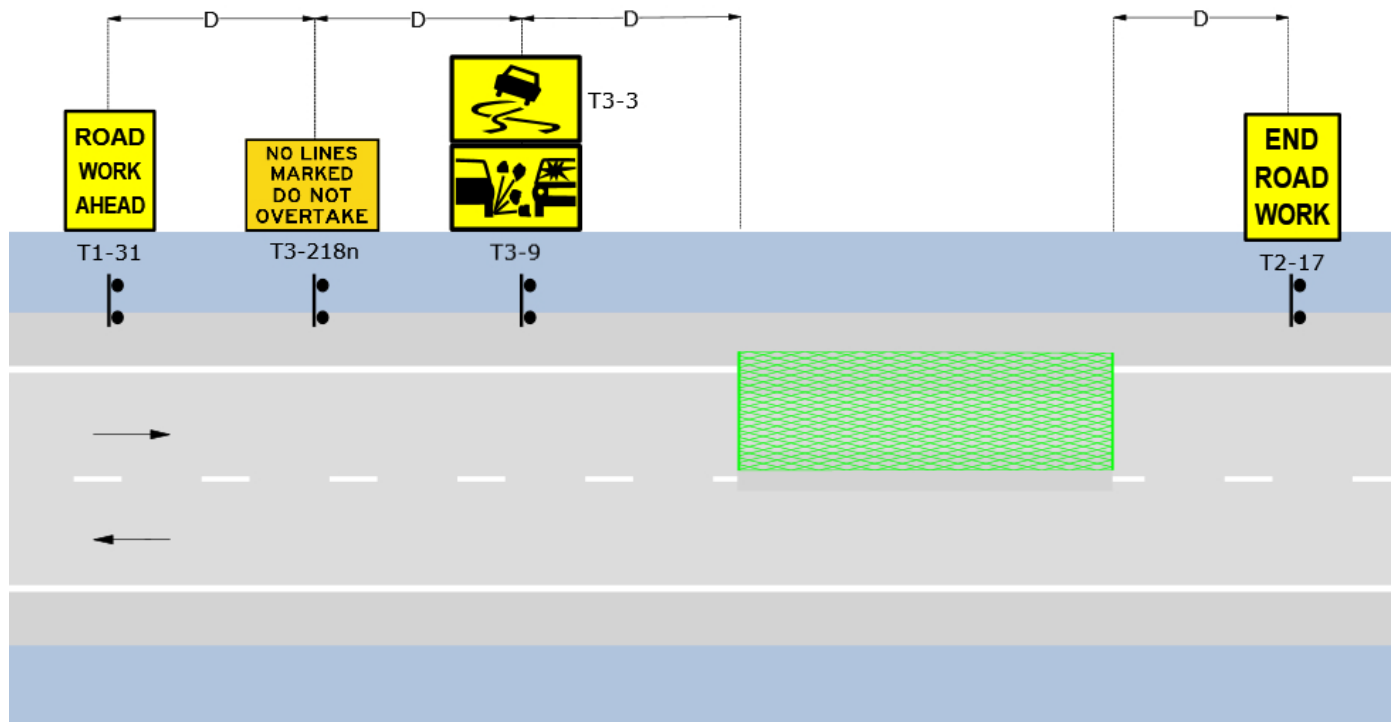


Figure 7-14. Example layout – bituminous works

7.7.5 Aftercare

Plans must be in place to ensure that traffic can proceed around, past or through a work site safely when workers are not present. This is typically referred to as *aftercare* traffic management. This includes periods outside normal working hours in either day or night when there are interruptions to the continuity of the work.

A separate TGS must be designed when aftercare is required at the work site, with the following applicable provisions (see [Figure 7-15](#) for an example layout):

- The TGS must include details of the requirements to manage traffic around, past or through the work site outside normal working hours or when workers are not present at the site. This includes the removal or covering of any signs that are not applicable, particularly temporary work site speed limits;
- Unless the risk assessment of the work site indicates otherwise, the work site speed limit should be returned to normal if safe or, if not possible, no more than 20 km/h below the existing permanent speed limit, having regard to the work site conditions (e.g. rough or slippery unsealed road surface, loose aggregate from bituminous works, or excavations close to the roadway); and
- Inspections of the site should be undertaken in accordance with [Section 8.1 Work site inspections, reviews and audits](#).

As part of preparing the work site to be left unattended, the actions shown in [Table 7-5](#) should be considered.

Table 7-5. Aftercare considerations and actions

Considerations	Actions
Size of site	Reduce the extents of the work site as much as possible by reducing the length of the work zone and removing/reducing any width restrictions.
Temporary speed limits	If temporary speed limits have been installed, the design should consider whether these need to remain in place or whether a higher speed zone or the existing permanent speed limit can be reinstated (any changes to a different temporary speed limit must be approved).
Unsafe conditions	Identification of potential unsafe conditions, e.g. loose material on the road surface, poor quality or confusing delineation, plant parked incorrectly, with mitigation measures implemented.
Signs	TTM signs need to remain in place and serviceable while the site is not attended. Check that all signs are ballasted for windy conditions, reasonably level, mounted at the correct height and positioned correctly.
Delineation	Check that all delineation devices are clean, positioned correctly, and adequately ballasted, e.g. double weighted bollards.

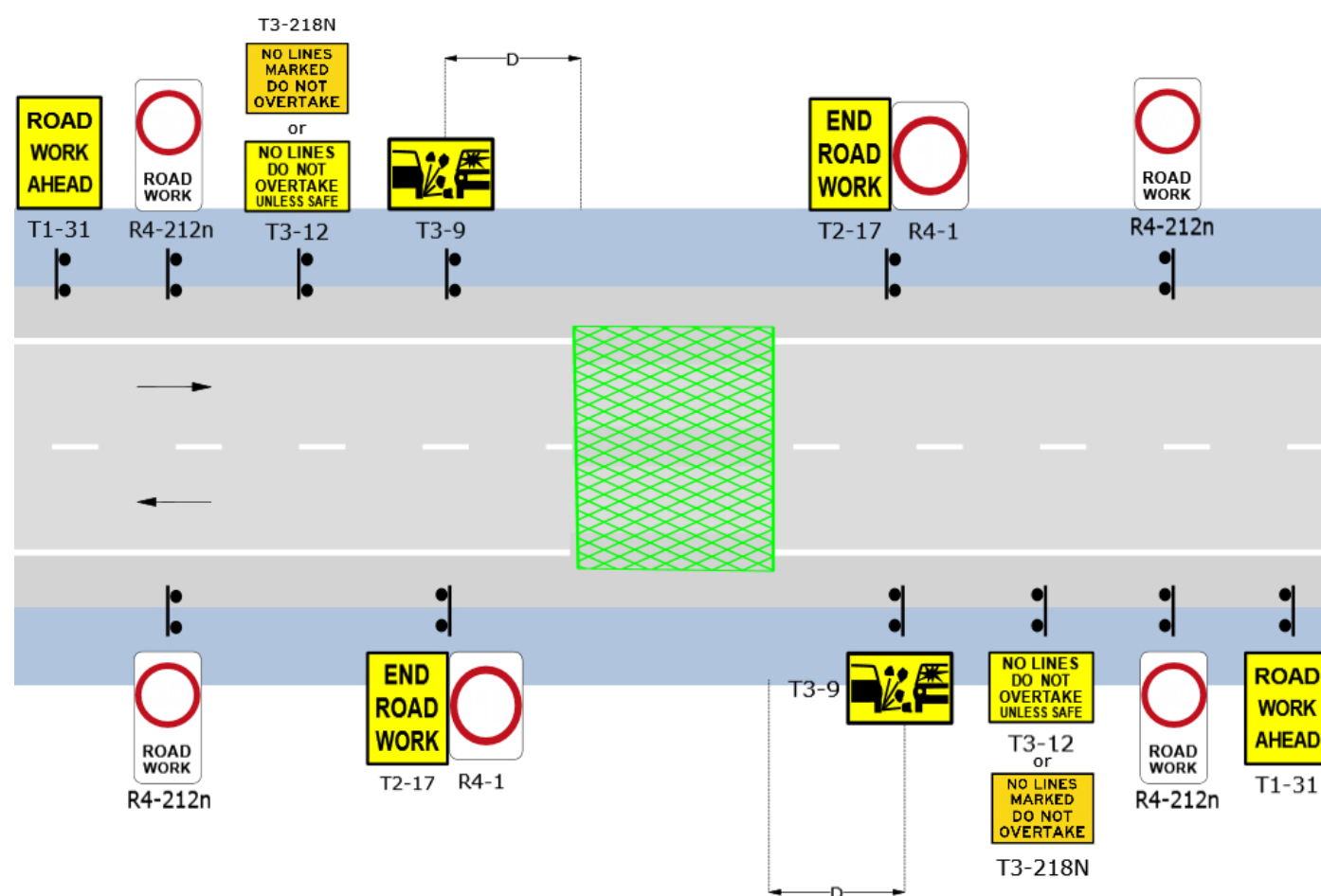


Figure 7-15. Example layout – aftercare

7.8 Designing a TGS: Dynamic work

7.8.1 General

Dynamic work involves the carrying out of a work activity as one of the following:

- **Frequently changing work** – regularly moves between successive locations, either within or outside of a traffic lane where minimal warning is required to advise road users of the presence of workers;
- **Continuous work** – progressively moving vehicles along the roadway, often either wholly within, or wholly outside of a traffic lane; or
- **Intermittent work** – work which is undertaken on travel lanes, in gaps in traffic, and requires no adjustment that affects road users on the roadway.

The general principles that must be applied when developing a TGS for dynamic works are provided in [Table 7-6](#).

Table 7-6. Principles for the design of a TGS for dynamic works

	TGS principles
Work site layout	• The work area should be designed to allow for movement of road workers, equipment, materials and vehicles, including sufficient waiting and storage space for TTM components. • The size and position of the work area should enable the work site to be managed effectively for the safety of road workers and road users. • The size and position may change during works depending on project tasks or location.
Sight distance	Sight distance allows a driver to see the work, react, and take required action to avoid a collision. Sight distance must: • Be determined in accordance with Section 4.3.6 Sight distances; and • Be appropriate for traffic composition (e.g. PBS vehicles see Section 4.4.5 Heavy vehicles)
Traffic volume	• Predicted end-of-queue lengths must be determined and a strategy must be documented in the TGS when higher than anticipated traffic volumes occur. • Unless unavoidable and essential, work should not be undertaken during peak times or special events that can influence traffic volumes.
Road users	When undertaking dynamic work, access provision must be maintained for impacted road users, e.g. users of foot paths or bicycle lanes must have access to safe alternatives
Speed	• The TGS must ensure that the appropriate speed zoning is in place for the dynamic work to be performed safely. • If speeding vehicles are predicted or noticed in the area, additional consideration should be given to advanced warning signs.
Vehicle placement	Dynamic works should be planned so that work vehicles do not need to straddle an edge line, and the placement of the work vehicles does not put at risk vulnerable road users. Note: Some dynamic work such as linemarking or creation of a mobile taper may require lane straddling.
Access to medians and verges	• Safe access and egress from medians and verges must be determined when planning dynamic work. • The road alignment and potential for vehicles to run-off onto median and verge should be considered when planning dynamic work in the median or verge.
Contingency	When performing dynamic work, a contingency plan must be developed if work cannot be completed as planned.

7.8.2 Work convoys

Work convoys, as a whole or select convoy vehicles, might be required when undertaking dynamic work. The need for a convoy or otherwise is influenced by factors such as:

- Lane and shoulder widths;
- Geometric alignment – horizontal and vertical alignment;
- Sight distance;
- Grade of road;
- Traffic volumes;
- Speed of traffic;
- Number of traffic lanes;
- Traffic composition (i.e. proportion of heavy vehicles);
- Road user behaviour;
- Traffic movements – such as weaving traffic; and
- Workers on foot.

Where used, a works convoy must be made up of the vehicles in [Table 7-7](#), as applicable, for the relevant road situation ([Figure 7-16](#) shows a typical layout of a works convoy).

Table 7-7. Vehicles and their function in a convoy

Vehicle	Role and function
Lead vehicle	A 'lead vehicle' is required on two-way roadways when working on the dividing line. It precedes a work area and has three main purposes: • Warns oncoming road users of the work; • Alert following workers early to any impending danger from oncoming traffic; and • Shield workers and the work vehicle from approaching traffic.
Work vehicle	A 'work vehicle' is the vehicle or item of plant which undertakes the work, e.g. a linemarking machine, or supports the workers on foot behind it.
Shadow vehicle	A 'shadow vehicle' follows closely behind the work area as the work progresses. Its main purpose is to shield the work vehicle and/or workers on foot from traffic. This vehicle must travel at a clear distance of 20 m to 40 m behind the work vehicle and consideration should be given to fitting the vehicle with a truck mounted crash attenuator when it is protecting workers in a traffic lane.
Tail vehicle	A 'tail vehicle' follows some distance behind the work area. The main functions of a tail vehicle are to: • Warn following road users of the work; • Divert traffic past the work area; and • Enable the driver to alert workers of any impending danger. Note: A second tail vehicle is required on motorway type roads.
Advance warning vehicle	An 'advance warning vehicle' should be positioned approximately 1 km behind the work convoy to give advance warning to following road users. • All signs and warning devices must be mounted on moving vehicles in the convoy; and • All vehicles and plant items in the works convoy, other than minor plant items, must have a flashing arrow sign fitted. An advance warning vehicle is not required for speed zones of less than 65 km/h with 2D sight distance.

Note to Table 7-7: Additional requirements for vehicles in a convoy for motorway environments are provided in [Section 7.8.4.4 Additional requirements: Work on motorway type roads](#).

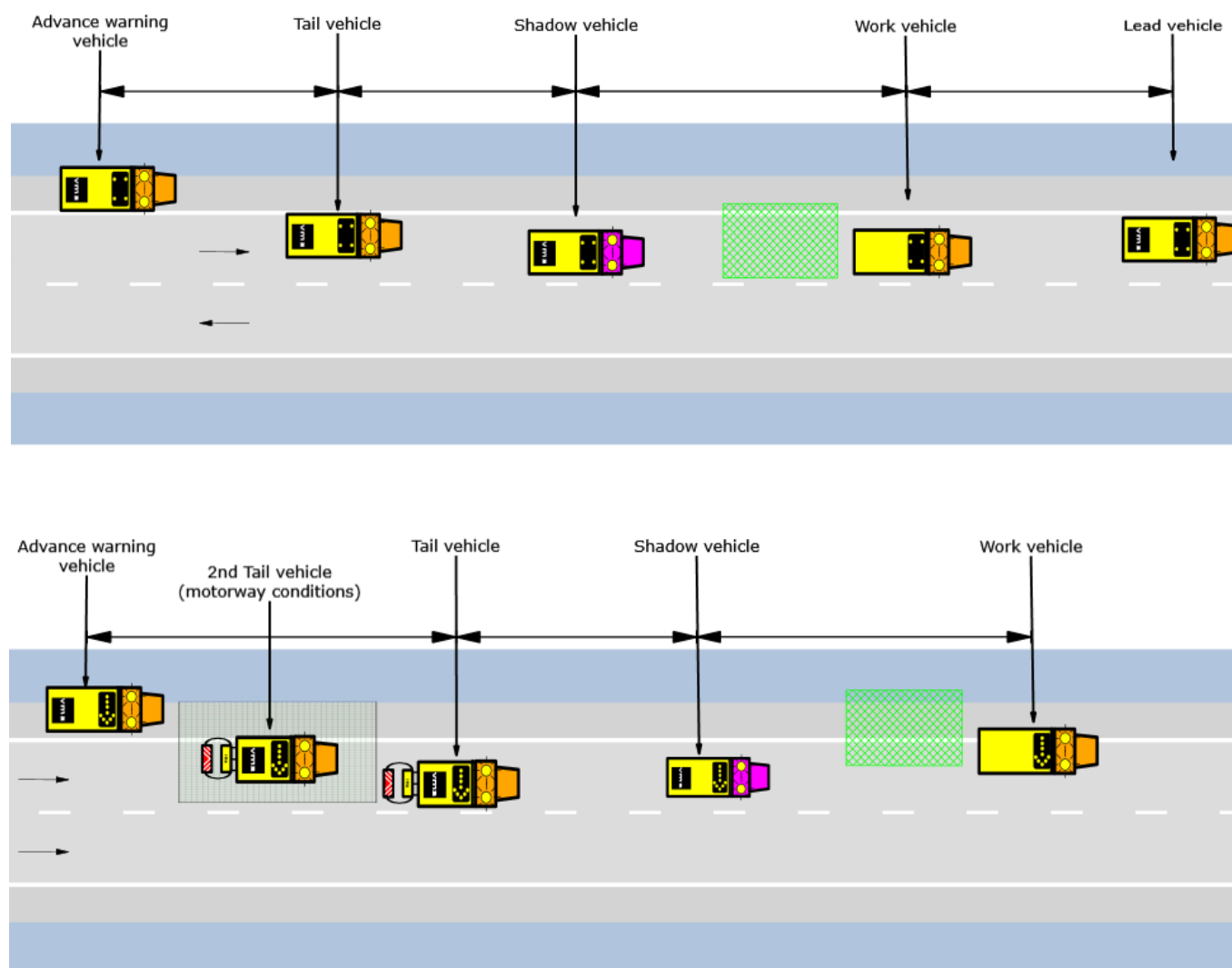


Figure 7-16. Works convoys – two-lane, two-way (top) and motorway conditions (bottom)

7.8.3 Frequently changing work

Frequently changing work involves regular movement between successive locations, either within or outside of a traffic lane where minimal warning is required to advise road users of the presence of workers. For frequently changing work to be conducted, the following criteria must be met:

- When conducted within a traffic lane:
 - Speed limit of less than 65 km/h; and
 - Maximum work period at any one location is 20 mins or up to 1 hour if traffic volumes are less than 40 vph.
- When conducted in a median or verge:
 - Sight distance that meets the criteria outlined in [Table 7-9](#) (or 50 m minimum sight distance to oncoming traffic not in adjacent traffic lane);
 - Speed limit of less than 75 km/h; and
 - Maximum work period at any one location with one of the following durations:
 - 1 hr at traffic volume less than 40 vph; or

- 40 mins at traffic volume 40 to 150 vph; or
- 20 mins at traffic volume more than 150 vph.

Where the above criteria cannot be met the site must be set up as a static work site or completed as another suitable dynamic work type.

Frequently changing work may permit the short term partial closure of a road without advanced warning where all of the following conditions are met:

- The vehicle-mounted warning device on the work vehicle can be seen by approaching traffic for at least 250 m;
- No traffic control is required; and
- Traffic volumes are less than 20 vpd or there is room for two-way traffic past the work area.

Examples include:

Work outside a traffic lane	Work within a traffic lane
• Pit cleaning or repair; • Litter collection; • Tree pruning or planting; • Road signs or street furniture maintenance; or • Street light maintenance.	• Pavement marker laying other than on dividing lines; or • Pavement testing.

7.8.3.1 Minimum controls for frequently changing work

When undertaking frequently changing work the following minimum controls apply:

- A shadow vehicle or works convoy must be deployed to protect worker on foot or plant;
- All work vehicles must have at least one operating rotating or flashing yellow light on the work vehicle when work is being completed or the work vehicle is parked that is:
 - Not obscured by overhanging vegetation or a raised truck body; and
 - Visible for at least 150 m in a speed zone less than 65 km/h or 250 m elsewhere.
- Sight distance that meets the requirements of [Table 7-9](#); and
- A contingency plan is in place so that work in progress can be abandoned immediately without risk to workers or traffic.

In addition, the following signage must be displayed:

- Signs up to 2 km in advance of each work position or item of moving plant. The distance between advance signs for opposing directions of travel must not exceed 2 km at any time and the location of such signs must be progressively changed to ensure the maximum separation is not exceeded as the work progresses along the road;
- At each advance sign location, either of the following signs must be used in conjunction with NEXT 2km (T1-28) and must be placed beside:
 - Workers symbolic (T1-5) where there are workers on foot; or
 - ROAD PLANT AHEAD (T1-3-1) where moving road plant only will be encountered.
- Where the work is repetitive and is undertaken within a designated length of up to 2 km, signs, such as WORKERS (symbolic) (T1-5) with NEXT 2KM (T1-28) or similar must be placed in advance of the work. Where this work is undertaken as a frequently changing work area, the advance signs must be progressively relocated so that in opposing directions they are never more than 2 km apart.

Where signs and devices are mounted on the vehicles or road plant they must be visible and conspicuous to road users at all times, see also [Figure 7-17](#).

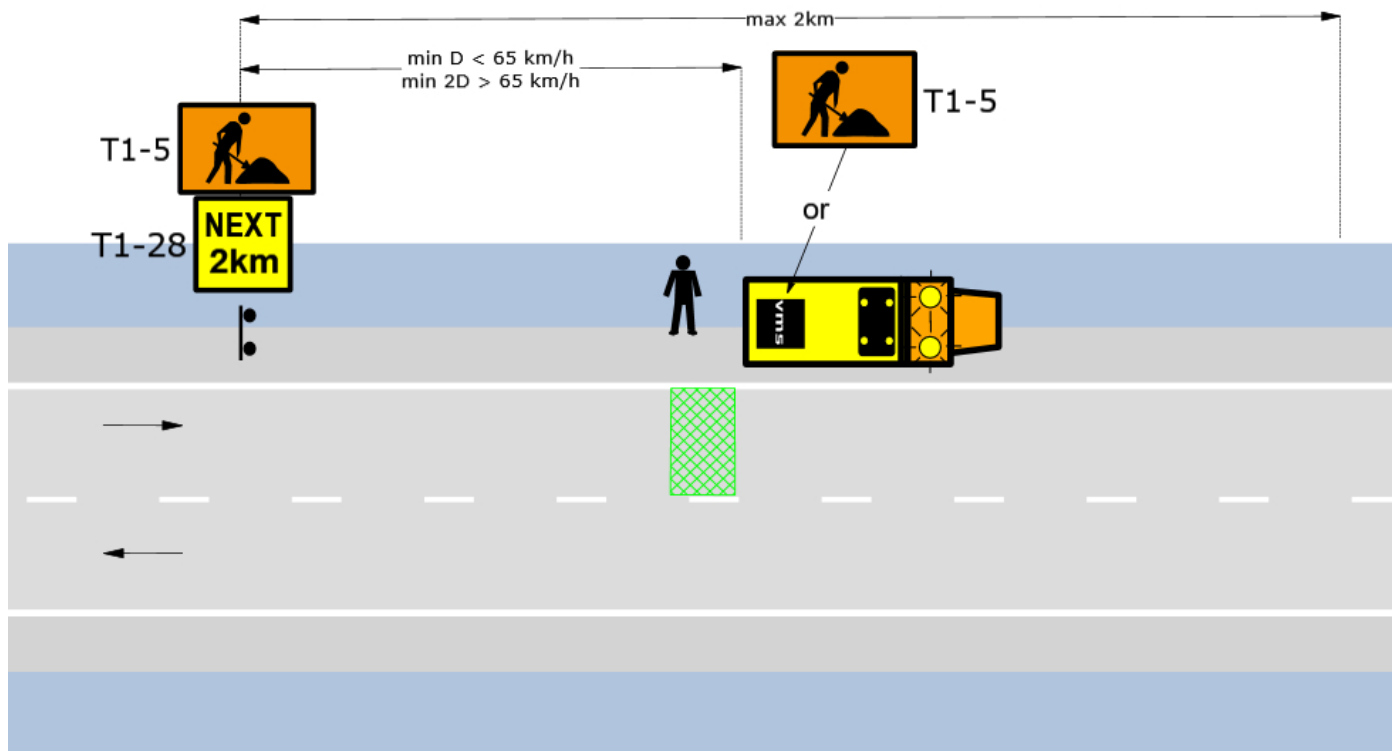


Figure 7-17. Example layout – frequently changing work

7.8.3.2 Additional requirements: Roads with less than 1500 vpd

Some works may be conducted as frequently changing work without the use of a works convoy or shadow vehicle provided all of the following conditions are met:

- Traffic volume is less than 1500 vpd;
- The minimum sight distance for following traffic is 1.5D;
- No more than two pieces of plant are operating within sight distance of each other;
- Posted speed limit is less than 85km/h; and
- At least 1.5m is maintained from live traffic.

Typically these activities include:

- Maintenance grading and resheeting of the traffic lanes on unsealed roads; or
- Shoulder grading on sealed or unsealed roads.

When undertaking frequently changing work without the use of a shadow vehicle, works may be undertaken within a road length of up to 10 km under the following conditions:

- Subsections of 2 km or less must be created and all signs must be installed before proceeding with the next section. If there is difficulty turning a grader around at the end of a 2 km section, it may be extended to the next available turning point but not to more than 6 km in total length;
- If sight distance to the vehicle-mounted warning sign on the works vehicle or plant is greater than 250 m throughout the section of road, ROADWORK NEXT __km (T1-24) must be placed at each end of the section.
- If the sight distance is less than 250 m at any point within the section:

- GRADER AHEAD (T1-4) or ROAD PLANT AHEAD (T1-3-1) together with NEXT 2km (T1-28) signs must be used on each approach to the section to cover the road length. These signs must be placed at least 100 m in advance of the start of any windrow; and
- If the speed of traffic is greater than 75 km/h, a temporary speed zone of less than 65 km/h must also be implemented, see [Section 4.5 Speed zones](#).
- Road condition signs must be placed at locations where the freshly graded surface has loose material that has become a hazard. Where required, one or more of the following must be installed depending on the nature of the hazard:
 - Slippery symbolic (T3-3);
 - SOFT EDGES (T3-6);
 - ROUGH SURFACE (T3-7);
 - GRAVEL ROAD (T3-13);
 - Loose stones symbolic (T3-9); or
 - LOOSE SURFACE (T3-14).

Additional information for specific activities are detailed in [Table 7-8](#), see also [Figure 7-18](#). Where traffic volumes are greater than 1500 vpd the works must be undertaken either as a continuously moving or static work.

Table 7-8. Additional information for specific activities

Activity	Additional information
Street sweeping	When undertaking works, the street sweeper must operate in a forward direction and not need to reverse for an extended distance. In addition, the sweeper may use a flashing arrow sign to direct traffic to one side of the vehicle if it is safe to do so.
Maintenance grading and resheeting	Wherever practicable, grading on the right side of the road against oncoming traffic should be avoided. Maintenance grading and resheeting may be carried out either with or without leaving a windrow. Work done without leaving a windrow normally involves use of either a windrow eliminator or a second grader in tandem. Where graded or resheeting material cannot be traversed by traffic, the grader blade should be raised from time to time to create a gap in the material or windrow so that traffic can overtake the grader.

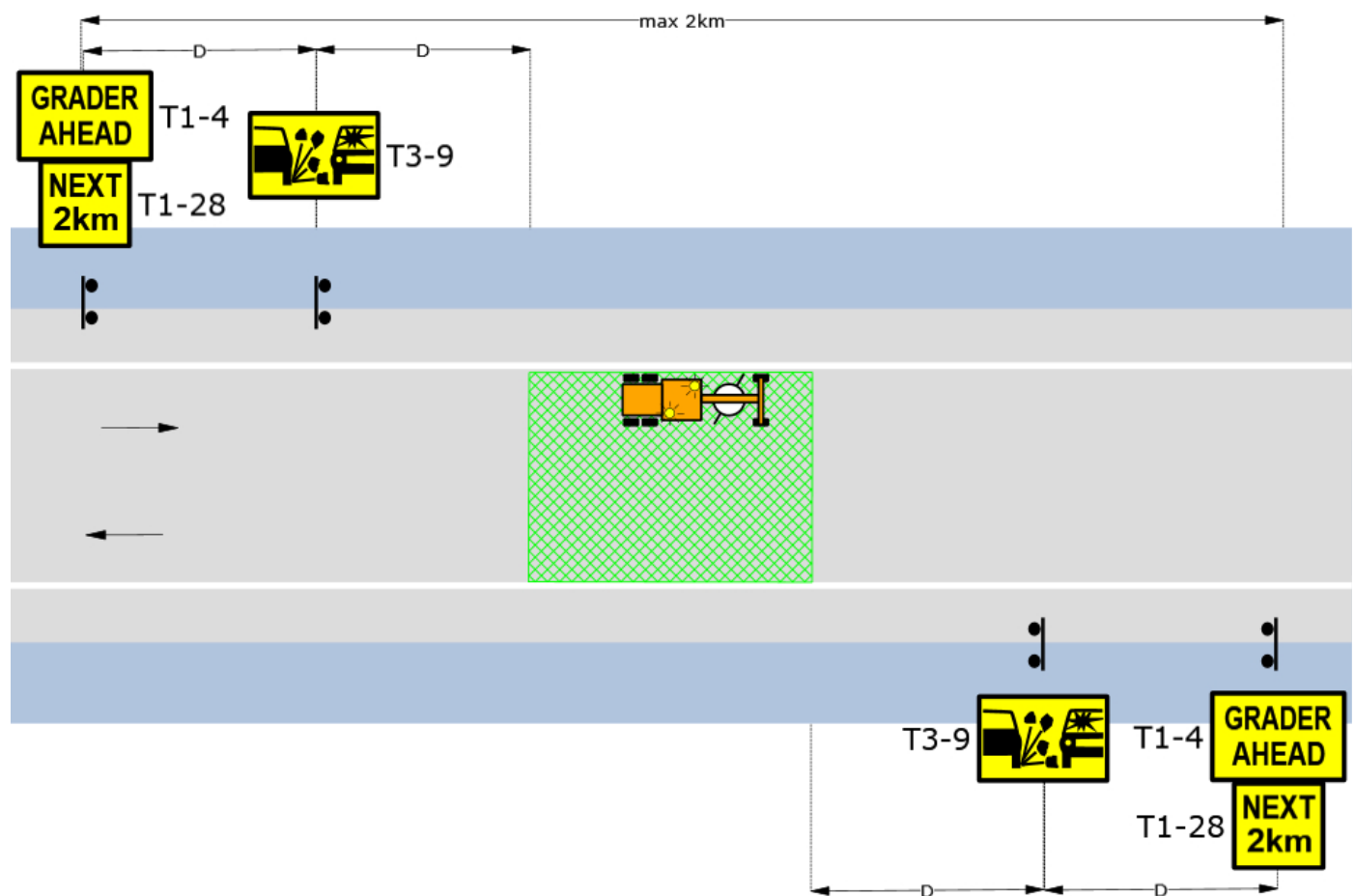


Figure 7-18. Example layout – grader ahead

7.8.3.3 Additional requirements: Work on a verge, median or footway

Access to verges, medians or footways must be carefully planned as it may involve works vehicles to manoeuvre in a manner that is unexpected by road users. Access to medians must be in accordance with [Section 5.2 Providing safe movements for works traffic](#).

Median and verge mowing, tilling, seeding and weed spraying, works on a footpath and garden maintenance may be carried out without a shadow vehicle, subject to a risk assessment, provided the work is short term work and all of the listed conditions being met:

- Where there are no workers on foot, the relationship between speed limit and clearance to edge of traffic lane must be as follows:
 - Clearance of at least 1.5 m for speed limits greater than 85 km/h; or
 - Clearance of less than 1.5 m are permitted for speed limits less than 85 km/h, however plant items must not encroach onto the traffic lane.
- Where there are workers on foot:
 - The clearances detailed in [Section 4.3.4 Minimum clearances of workers to traffic](#) must be maintained; and
 - Where works will require workers on foot at one location for periods longer than 20mins, cones or bollards, must be placed along the kerb line or edge of traffic lane if no kerb exists.

7.8.3.4 Additional requirements: Slow moving plant

Plant that moves along the roadway that cannot travel within 20 km/h of the posted speed limit may be moved using the principles of a 'frequently changing work site' in a traffic lane using one of the following actions:

- Put in place a TGS to stop traffic while the plant is moved. It would normally be necessary to stop traffic only in the direction the plant item is travelling;
- The plant item travels on the road shoulder or verge using gaps in traffic to pass obstructions and does not straddle the edge line; or
- Use a pilot vehicle or convoy of vehicles in accordance with the below.

Where required to pilot slow moving plant items travelling on the road, the following provisions apply:

- At least one pilot vehicle must travel behind the plant item;
- Up to three plant items may be guided in the one group using one or two pilot vehicles as prescribed above, provided a distance of 60 m to 80 m is maintained between the plant items;
- Where the sight distance is:
 - Less than 2D, the pilot must travel at the same speed as the plant item, and must be at a distance of approximately 2D behind the plant item;
 - Between 1D and 2D, the pilot must stop on the shoulder and must be completely clear of the travel lane until the plant item has moved a maximum of 6D ahead. The pilot may then travel at normal speeds to a point near the plant item and stop on the shoulder once again;
 - Less than 1D, a second pilot vehicle must be used travelling at varying distances behind the plant item. The second vehicle will vary speed to ensure as far as possible that maximum sight distance is maintained for traffic approaching from behind. For example, the pilot would travel more quickly through a bend and then slow down on a straight where sight distance is increased. This vehicle should occasionally pull over to allow traffic to pass. The first pilot vehicle must continue to operate as described above.
- The plant items must pull over periodically to allow traffic to pass so that the queue of traffic following does not exceed 12 vehicles.

In addition to the requirements of [Section 7.6.5 Pilot vehicles](#), the PILOT VEHICLE DO NOT OVERTAKE (T6-5) sign must be replaced with ROAD PLANT AHEAD (T1-3-1) sign.

Where the above conditions cannot be met, plant should be floated between sites.

7.8.4 Continuous work

Continuous work is defined as work which entails vehicles moving along the roadway continually. Continuous work may be conducted with plant only, or with workers on foot, but must be completed either wholly within, or wholly outside of a traffic lane.

Examples include:

Plant only activities	Worker on foot activities
• Linemarking; • Pavement testing; • Pavement sweeping; • Low speed inspections • Snow clearing/spreading grit;	• Linemarking removal; • Pavement edge or pot hole patching; or • Raised pavement marker installations.

Plant only activities	Worker on foot activities
- Grading - shoulder or unsealed road; or - Vegetation control.	

7.8.4.1 Minimum controls for continuous work

When undertaking continuous work (see [Figure 7-19](#)), the following minimum controls apply:

- All work vehicles must have at least one operating rotating or flashing yellow light on the work vehicle when work is being completed or the work vehicle is parked that is:
 - Not obscured by overhanging vegetation or a raised truck body; and
 - Visible for at least 150 m in a speed zone less than 65 km/h or 250 m elsewhere.
- All vehicles must stay wholly within or wholly outside of a travel lane; and
- If travelling more than 20 km/h below the posted speed limit, be protected by a works convoy as detailed in [Section 7.8.2 Work convoys](#).

In addition, vehicles may:

- Use a flashing arrow sign to direct traffic to one side of the vehicle provided it is safe to do so. In situations where it is not safe to pass or overtake the work vehicle, only the warning mode of the arrow must be flashed; and
- Require supplementary vehicle mounted signs such as:
 - WET PAINT ON ROAD (T2- 237n) or
 - ROAD PLANT AHEAD (T1-3-2).

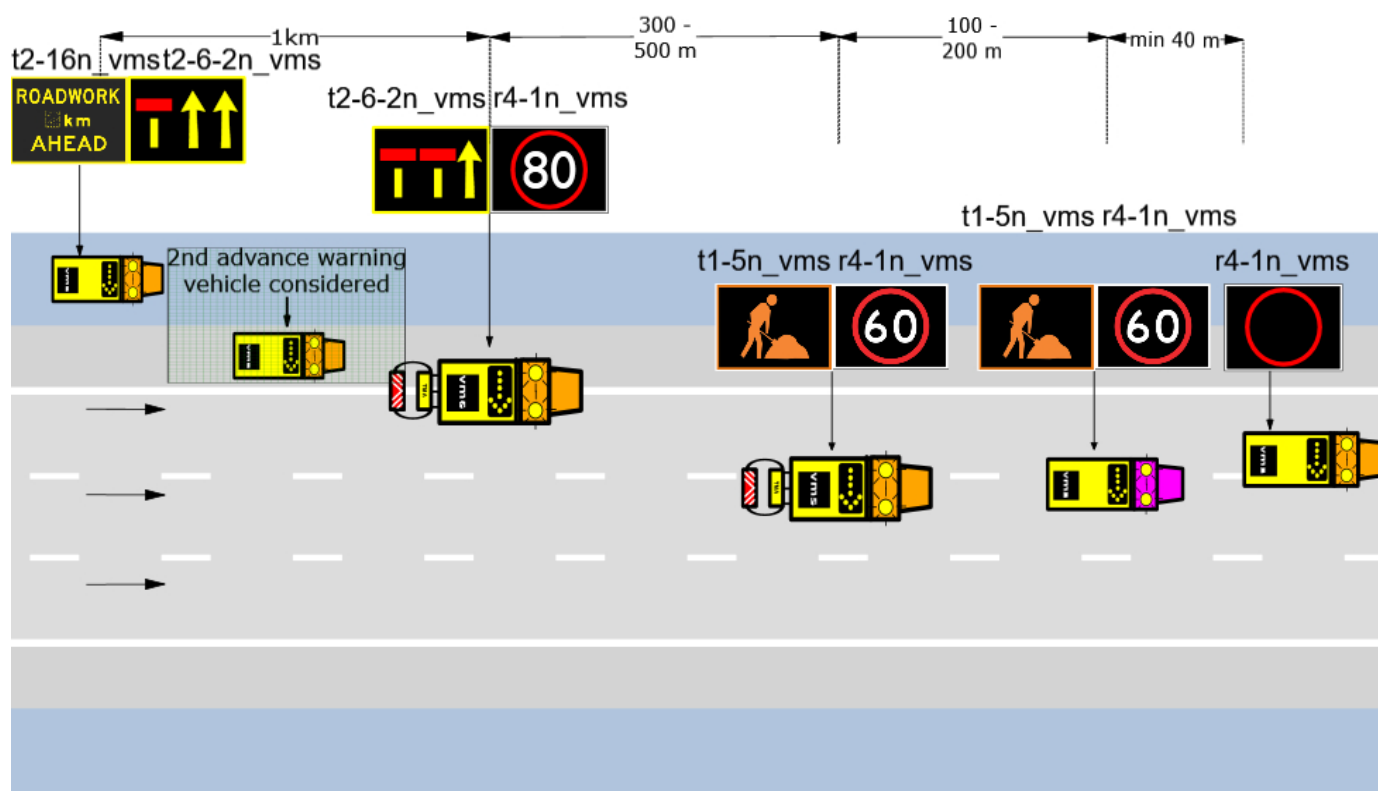


Figure 7-19. Example layout – continuous work

7.8.4.2 Additional requirements: Advanced warning

Advance warning vehicles should be positioned in the same lane as the work vehicle. However, for work near the centre of a multi-lane undivided road or near the median on a divided road, the advance warning vehicle should be to the right of the carriageway to divert traffic around the left of the work area.

Vehicles providing advance warning must travel at the following convoy spacing:

- Where sight distance is greater than 2D:
 - Lead vehicles travel 200 m to 400 m in open road areas or 30 m to 100 m in built-up areas in advance of the work vehicle; and
 - Tail vehicles travel 300 m to 500 m in open road areas or 200 m to 300 m in built up areas behind the work vehicle or shadow vehicle if one is being used. This vehicle may be dispensed with if the speed limit is less than 65 km/h.
- Where sight distance is less than 2D:
 - The lead vehicle must move as necessary beyond the distances given above to a point where sight distance of 2D is regained, and remain there until the work vehicle catches up; and
 - The tail vehicle must hold at a position of 2D sight distance until the work vehicle has progressed to a point where the tail vehicle can move through the section with restricted sight to a point where 2D sight distance is regained.

7.8.4.3 Additional requirements: Work on two-way roads

Wherever practicable, following traffic must be directed to pass to the left of the work convoy by flashing the left barb of the flashing arrow sign.

Where it is not possible for traffic to pass to the left and opportunities to safely pass to the right do not exist the work must be stopped periodically by pulling off the traffic lane and allowing following traffic to pass.

Traffic must not be directed to completely cross a dividing line into the path of oncoming traffic. Additionally, traffic must not be directed to cross a freshly marked line if that would result in the marking being damaged.

7.8.4.4 Additional requirements: Work on motorway type roads

The shadow vehicle (second vehicle) in the convoy or any other vehicles occupying a travel lane (or part thereof) in a motorway environment must be fitted with a truck mounted crash attenuator (TMA), see [Figure 7-16](#) for further details. See [Section 6.10 Truck and trailer-mounted attenuators \(TMA\)](#) for specific requirements for the operation of a TMA.

A second tail vehicle must be provided for work on motorway type roads. The two vehicles must be arranged to form a mobile taper with the first vehicle travelling to the left or the right of the occupied lane and the second (closer to the work vehicle) travelling in the occupied lane.

7.8.5 Intermittent work

Intermittent work does not require any adjustments that affect road users on the roadway. For intermittent work to be performed in a live traffic lane, work must be performed within safe gaps in traffic.

Intermittent work is only permitted as a work type when all of the following criteria are met:

- Sight distances in [Table 7-9](#) are achieved;
- Work can be performed within gaps in traffic; and
- Traffic is not obstructed.

The duration of intermittent work must not exceed:

- **5 minutes** – if works are within 1.5 m of the live traffic lane;
- **20 minutes** – if works are in the shoulder or greater than 1.5 m from the live traffic lane; or
- **60 minutes** – if works are wholly within a verge, median or footway.

Intermittent work must not include:

- Speed limit changes;
- Tapers;
- Traffic control, i.e. use of traffic control devices or manual traffic controllers;
- Aftercare signs;
- Redirection of vulnerable road users; or
- Lane width or alignment adjustments.

Work must be treated as a frequently changing or static work areas where there will be two or more intermittent locations within a space of 2 km or less worked simultaneously.

Examples of intermittent work are pot hole patching, edge repair, seal or patch spotting, road inspections, removal of dead animals or debris, or litter collection etc., see [Figure 7-20](#).

In addition to the TGS requirements in [Section 7.4 Information required on a TGS](#), where applicable, when designing a TGS for intermittent work, the following information must be provided:

- A contingency plan so that work in progress can be abandoned immediately without risk to workers or traffic;
- Sight distances available for the work location; and
- Whether a lookout person is required to warn workers on foot of the approach of any vehicles whose speed or size might constitute a safety threat.

7.8.5.1 Minimum controls for intermittent work

When undertaking intermittent work, the following minimum controls apply:

- All work vehicles must have at least one operating rotating or flashing yellow light on the work vehicle when work is being completed or the work vehicle is parked that is:
 - Not obscured by overhanging vegetation or a raised truck body; and
 - Visible for at least 150 m in a speed zone less than 65 km/h or 250 m elsewhere.
- All road workers, materials and plant have the ability to quickly and safely move on or off the roadway in a short period;
- A contingency plan, so that work in progress can be abandoned immediately without risk to workers or traffic;
- Sight distances outlined in [Section 7.8.5.2 Sight distances for intermittent work](#); and
- Where works involve workers on foot or small items of plant working within 3 m of a traffic lane, the Worker symbolic (T1-5) sign and/or ROAD PLANT AHEAD (T1-3-1) sign must be displayed.

When work is undertaken with a lookout person, the following provisions apply:

- There must be no approaching traffic for at least a minimum distance as shown in [Table 7-9](#) for the designated approach speed, i.e. approximately equivalent to 10 seconds travel time or 3D;
- The lookout person must have good eyesight and must be fully instructed in the lookout persons' duties; and

- The lookout person must remain within sight and hearing distance of the worker(s) (generally within 2 m).

Where the distances shown in *Table 7-9* cannot be achieved, an advance lookout person should be used. This advance lookout person should be within sight distance of the first lookout person or within radio contact if sight distance is not possible.

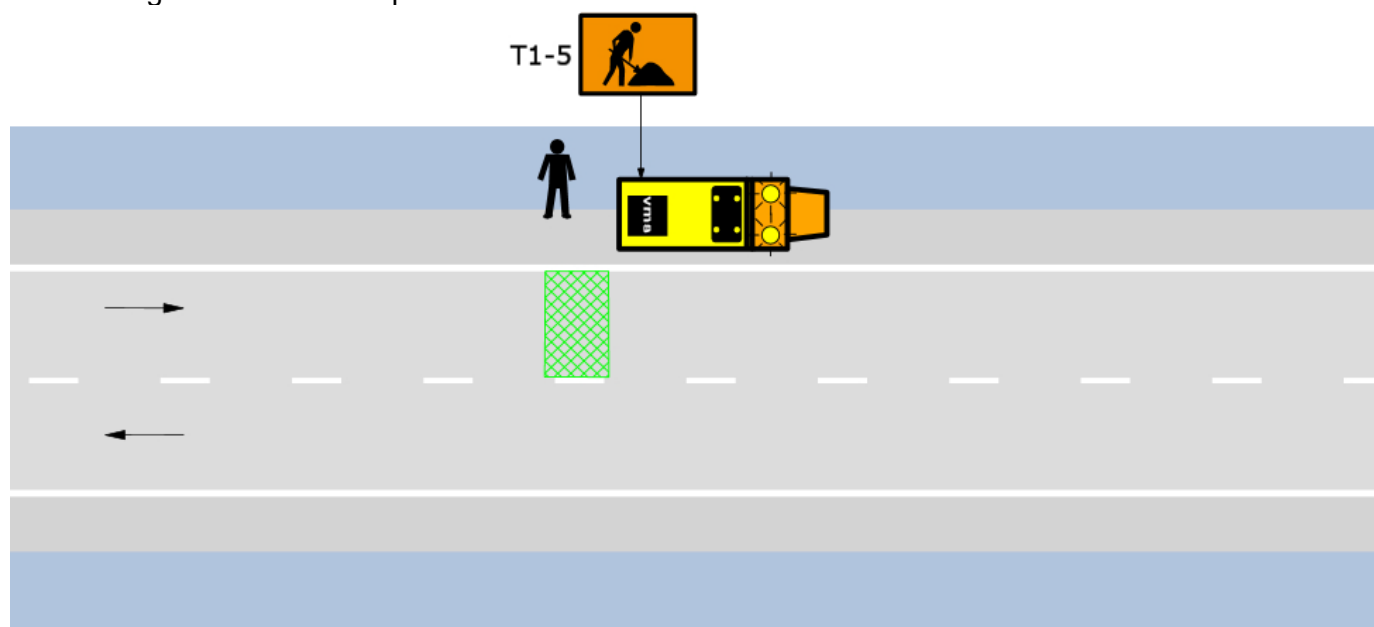


Figure 7-20. Intermittent work

7.8.5.2 Sight distances for intermittent work

When undertaking intermittent work, the need for a lookout person will be dependent on the sight distance available at the location. Sight distances, detailed in *Table 7-9* must be maintained at all times.

Table 7-9. Minimum sight distances for intermittent work

Approach speed (D) km/h	Minimum sight distance (m)	
	With a lookout person (3D)	Without a lookout person (6D)
20 or less	60	120
30	90	180
40	120	240
50	150	300
60	180	360
70	210	420

Approach speed (D) km/h	Minimum sight distance (m)	
	With a lookout person (3D)	Without a lookout person (6D)
80	240	480
90	270	540
100	300	600
110	330	660

7.9 TGS confirmation and approval

7.9.1 TGS verification

After a TGS is selected or designed it must be approved as being appropriate for use at the work site. Site confirmation must be undertaken via the completion of the TGS verification.

A TGS verification must be undertaken to confirm the selected or designed TGS is fit for purpose. A TGS verification must be completed in accordance with Section 8.1.2 TGS verification by an ITCP or PWZTMP qualified person. TGS verification must include an inspection of the work site where the TGS will be implemented.

7.9.2 TGS approval

The PWZTMP qualified person who has designed or modified the relevant TGS must approve the TGS for use. Approval of the TGS must include:

- Review of the relevant TMP, risk assessment and associated TTM specific documentation;
- Design, redesign or modification of the TGS must be in accordance with the requirements of this Technical Manual; and
- Confirmation that the TGS provides the relevant information for the ITCP person to safely implement on-site.

The PWZTMP qualified person must seek approval from the one up manager if the TGS:

- Includes a non-standard or unaccepted sign or device;
- Includes any departures from the requirements of this manual; or
- If a manual traffic controller is proposed for use.

The TGS must include separate sign-off sections for the designer and approver. The sign-off sections must include the relevant persons:

- Full name;
- Role;
- Division / organisation;

- Qualification number;
- Signature; and
- Date.

7.10 Implementing a TGS

7.10.1 General

A TGS must be installed, maintained and removed in a planned and safe manner. The implementation of a TGS must only be undertaken by an ITCP qualified person.

Before work begins, the ITCP qualified person must confirm that:

- The TGS has been selected, designed or modified and approved in accordance with this Technical Manual;
- Appropriate approvals have been given for road occupancy, speed zone authorisation, PTCD or use of manual traffic controllers;
- Copies of the relevant TMP, associated risks assessments and other relevant documents are obtained;
- Adequate materials (i.e. sign covers) are available to cover signs that are not needed; and
- All signs and devices on the TGS are available, are of correct size and are in good condition. Sufficient additional devices should be available to allow for any on-site adjustments, modifications and/or contingencies including:
 - Additional cones;
 - Repeater signs;
 - Additional signs for end-of-queue management;
 - Covering of existing conflicting signage;
 - Covering of signs not needed for aftercare TGS; and
 - Incident/emergency signage.

Note: Modifications and adjustments to a TGS must be undertaken by an appropriately qualified person and must be in accordance with this Technical Manual.

7.10.2 Implementing a TGS

The sequence of implementation should be determined as part of the TMP, and detailed on the TGS or SWMS, rather than being determined on-site.

The general procedure for setting up a site should be in accordance with Section 6.4 Sequence for installation and removal of signs and devices. A different implementation sequence may need to be adopted to address any site specific circumstances.

The ITCP qualified person must ensure that the TGS is implemented as approved, with any minor adjustments completed in accordance with Section 7.10.3 Tolerances on positioning of signs and devices, are recorded on the TGS and a signed copy is available on-site.

Any anomalies or inconsistencies found in the TGS being used must be recorded and reported back to the TGS designer who is PWZTMP qualified.

If it is identified that risk will be generated through the implementation of a TGS that requires modification outside of the tolerances, then the site must be made safe and an updated TGS must be provided by a

PWZTMP qualified person as soon as practicable. Where the TMP and TGS cannot be suitably adjusted or modified, the ITCP qualified person must advise the project manager that they are not appropriate and the works be postponed until a new site specific TGS is designed.

Once implemented, the appropriate inspection in accordance with Section 8.1 Work site inspections, reviews and audits must be completed.

7.10.3 Tolerances on positioning of signs and devices

Local constraints might not allow signs and devices to be placed exactly in accordance with the designed and approved TGS.

Where a specific distance is provided for the longitudinal positioning of signs or devices with respect to other items or features, the tolerances provided in Table 7-10 may be applied.

Table 7-10. Permitted tolerances for positioning of signs and devices

Tolerance	Positioning of signs, length of tapers or markings	Spacing of delineating devices
Minimum	10% less than the distances or lengths given	Nil
Maximum	25% more than the distances or lengths given	10% more than the spacing shown

These tolerances must not be applied if a maximum, a minimum or a specific range has been specified on the TGS. They may need to be exceeded where road features such as intersections or median openings occur, provided approval is granted by the PWZTMP qualified TGS designer.

Any sign location changes must be marked and initialled on the TGS held on site, with the name of the person making the changes shown.

7.10.4 Modifying a TGS

Modifications to a Site Specific TGS must be approved by a person holding the PWZTMP qualification, and must be supported by a TMP or risk assessment to ensure all TGSs consider and mitigate identified site-specific conditions and risks.

The ITCP qualified person may vary the positioning of signs and devices provided the requirements of Section 7.10.3 Tolerances on positioning of signs and devices are met.

If it is identified that risk will be generated through the implementation of a TGS that requires modification outside of the tolerances, then works must be stopped (including the implementation of the TGS), the site must be made safe and an updated TGS must be provided by a PWZTMP qualified person prior to works recommencing.

7.10.5 Monitoring activities

In accordance with Section 8.1 Work site inspections, reviews and audits, initial and regular inspections must be completed.

Appendix E – Inspection checklists and tools, provides a checklist that may be appropriate for this purpose.

7.11 TGS review

7.11.1 Generic TGS review

Generic TGSs must be reviewed by a PWZTMP qualified person every 12 months so that they remain appropriate. Once reviewed the date and details of the PWZTMP person must be updated on the TGS to ensure persons selecting can confirm currency.

7.11.2 Site Suitable and Site specific TGS review

All active Site Specific and Site Suitable TGSs must be reviewed as part of the weekly inspections detailed in Section 8.1 Work site inspections, reviews and audits.

A Site Specific or Site Suitable TGS must only be used for the duration of the work activity noted on the TGS. If the work activity is intended to be longer than 12 months, then the TGS must be formally reviewed by a PWZTMP qualified person at least every 12 months and issued with the review date and the details of the PWZTMP qualified person undertaking the review, and recorded on the TGS.

7.12 TGS records

All implemented TGSs must be recorded and stored in accordance with Section 8.2 Record keeping of TTM documentation.

8 Work site inspections, recording and reporting

8.1 Work site inspections, reviews and audits

8.1.1 General

The inspection, review and audit of temporary traffic management (TTM) arrangements are critical to ensure that the work site is operating safely. As such, the structure, schedule and frequency of these activities must be considered and identified during the TTM planning phase. These aspects will vary depending on the size, complexity and duration of works.

There are two categories of monitoring activities that occur during TTM:

- **Mandatory monitoring activities** – these are required for all TTM arrangements (summarised in [Table 8-1](#)); and
- **Additional monitoring activities** – these are provided to assist the TTM application and may be prescribed as mandatory by Transport as divisional process in the relevant G10 specification or undertaken on behalf of client representatives as part of other functions (summarised in [Table 8-2](#)).

TTM monitoring activities must be undertaken in all instances where work is being performed or aftercare is in place. This includes during the day and at night times.

Table 8-1. Mandatory monitoring activities during TTM

Stage	Activity	Purpose	Qualification	Tools and checklists
Planning	TGS verification	To ensure that the TGS selected or designed is suitable for the works and location.	ITCP or PWZTMP	Appendix E.2 TGS verification checklist
During TTM	Weekly TTM inspections (includes preopening inspection)	To ensure that the TMP and relevant TGS are appropriate and operating safely, effectively and efficiently	PWZTMP	Appendix E.3 Weekly TTM inspection checklist
	Shift TTM inspections	To ensure that the TGS is implemented as designed. This includes at a minimum, twice per shift and when: • A TGS is installed, changed or updated; • At regular frequency after work commences, recommended every 2 hours; and • Once aftercare arrangements have been installed if required	ITCP or PWZTMP	Appendix E.4 Shift / Daily TTM inspection checklist
	TMP review	To ensure that TMP controls are achieving the required outcomes.	PWZTMP	Not provided
Post completion	Post-completion inspection	To ensure that the site has been demobilised as planned and is safe for opening to traffic	ITCP or PWZTMP	Appendix E.5 Post completion inspection checklist

Table 8-2. Additional monitoring activities during TTM

Stage	Activity	Purpose	Qualification	Tools and checklists
During TTM	TTM Road Safety Audits	To identify road safety crash potential and areas of risk that could lead to traffic incidents in relation to the application of TTM. The aim of a road safety audit is undertake an objective assessment of crash risk.	Registered Road Safety Auditor and PWZTMP	Not provided
	Client inspections	Verification of TTM through the Transport Traffic Engineering, Work Health and Safety Branch, Surveillance Officers or other client representatives.	Divisionally determined	Not provided

8.1.2 TGS verification

In accordance with [Section 7.9 TGS confirmation and approval](#), prior to use on site a TGS must be verified to ensure it is suitable for the works and the location. An example TGS verification checklist has been included in [Appendix E – Inspection checklists and tools](#) that may be suitable for this purpose.

Where a physical site inspection cannot be completed due to extenuating circumstances, such as emergency works or excessive distance, the TGS verification may be completed via the collection of the relevant information from other sources. Alternative sources of relevant information include aerial or street view images.

The purpose of the TGS verification is to ensure that the TGS selected or designed identifies and manages the site-specific risks associated with the location. As such the TGS verification must validate the accuracy of the data and information provided in the TMP and TGS, including but not limited to the existing permanent speed limits, and identifying any other items that may generate a risk or hazard during the works that have not already been considered, such as bus stops or parking.

Information that should be considered when undertaking a TGS verification includes:

- Shoulder widths;
- Sight distances;
- Existing permanent speed limit;
- Existing infrastructure such as signs, barriers, buildings and utilities;
- Transport services such as bus stops or parking facilities; and
- Other sources of risk for the works, such as community facilities.

Once completed, the TGS selector or designer, must ensure that all relevant supporting information, including records of verification, are included with the TGS for implementation.

8.1.3 Weekly TTM inspections

After a TMP and TGS have been approved and the TTM work site is established in accordance with this Section, a weekly TTM inspection must be completed prior to the initial opening of a work site, regardless of duration of work.

Note: This means that the TTM has been installed, but the work intended to be protected by the TTM has not yet commenced.

After the initial opening inspection, the same inspection must be completed on a weekly basis thereafter. Appendix E – Inspection checklists and tools provides a checklist that may be suitable for carrying out weekly TTM inspections.

The weekly TTM inspections must focus on effectiveness of the TMP in the context of the work being performed and that the TGS has been appropriately implemented.

The weekly TTM inspection must be:

- Undertaken by a person holding the PWZTMP qualification;
- Carried out from within the traffic stream at the normal traffic speed;
- Undertaken:
 - After the TGS has been implemented but prior to the work commencing on-site for the first time, known as a pre-opening inspection; and
 - At least once per week thereafter, known as a weekly inspection.

The pre-opening inspection must be carried out after the required TGS is installed but prior to the intended work activity commences or prior to opening a switch to traffic to ensure that:

- The work site is established in accordance with the TMP and relevant TGS;
- The TMP and relevant TGS are:
 - Provided and are on site;
 - Approved; and
 - Implemented as prescribed.
- Safe passage has been provided for all road users to travel around, past or through the work site;
- Signs and devices are in good condition and clearly visible to road users; and
- Any potential hazards are identified and addressed in the TMP and TGS prior to opening.

Thereafter, the weekly TTM inspection must ensure all of the above are maintained in addition to verifying:

- The work site is operating safely and as intended, including risk identification and mitigation;
- All incidents and near misses are reviewed; and
- Inspections are being completed.

When undertaking a pre-opening or weekly TTM inspection, the person completing the inspection should:

- Have a copy of the approved TMP and TGS used on-site;
- Ensure inspections are completed in day and/or night conditions as required; and
- If inspecting during low light, ensure low beam headlights are used for the inspection.

8.1.4 Shift TTM inspections

After the work site has been opened, shift TTM inspections, also known as 'daily' inspections, must be undertaken on a regular basis to confirm that the site traffic management:

- Has been implemented in accordance with the approved TGS;
- Is operating as intended; and

- The controls implemented, are effective.

These shift TTM inspections ensure all road users are safely travelling around, past or through the work site. If vulnerable road user paths have been impacted, such as footways and cycle ways, the shift TTM inspection must include a walk-through or cycle-through as appropriate. The inspections must also include the surrounding road network if detour routes have been implemented.

The frequency of shift TTM inspections must be determined during the TGS development and approval phase by the PWZTMP qualified person, and recorded on the relevant TGS.

Shift TTM inspections must be completed at regular intervals throughout the work shift, at a minimum, shift TTM inspection must be completed twice per shift, however it is recommended that they be completed every two hours. TTM shift inspections must additionally be completed when:

- A TGS is installed, changed or updated; and
- Once aftercare arrangements have been installed, if required.

Shift TTM inspections must be undertaken by an ITCP or PWZTMP qualified person. When undertaking a shift TTM inspection, the person completing the inspection should:

- Have a copy of the TMP and TGS used on site;
- Carry out the inspection from within the traffic stream at the normal traffic speed;
- Ensure inspections are completed in day and/or night conditions as required; and
- If inspecting during low light, ensure low beam headlights are used for the inspection.

Where relevant, the shift TTM inspection must ensure that:

- Signs and devices:
 - Are installed and spaced in accordance with approved TGS;
 - Are in good condition, undamaged, in place and clearly visible to road users; and
 - Have been added, adjusted or moved in accordance with the approved TGS.
- There are no conflicting signs and devices;
- Delineation is clear and appropriate;
- Any variations, changes or departures to the TGS are recorded, approved and incorporated;
- Changed conditions do not impact on visibility of signs and devices (e.g. shade, parked vehicles etc.);
- There will be no drainage issues during a rain event from the positioning of signs or devices, or other safety issues arising from where traffic will be travelling through the work site;

In addition, a shift TTM inspection must be undertaken at least once for aftercare conditions to ensure:

- The relevant signs and devices have been covered and/or removed;
- The covers adequately cover the out-of-service sign;
- All covers are securely installed and will remain in place during windy conditions; and
- The appropriate speed limit is reinstated as per TMP & TGS.

A shift TTM inspection checklist is included in [Appendix E – Inspection checklists and tools](#) and may be used to keep a record of the inspection.

8.1.5 TMP reviews

TMP reviews are undertaken to ensure TMP elements are implemented as designed and remain appropriate. The TMP review must be undertaken by PWZTMP qualified person, who is independent to the installation of the TTM, usually the relevant project manager.

The objectives of a TMP review are to:

- Ensure that due consideration has been given to traffic management planning, risk identification and mitigation;
- Ensure that the work site is operating safely;
- Assess site conditions to ensure implemented speed limit is appropriate;
- Ensure that the TGS has been provided, is available on the work site, has been approved and has been implemented as prescribed;
- Ensure any variations to the TGS (for example in sign location due to shade, parked vehicles) are in accordance with Section 7.10.3 Tolerances on positioning of signs and devices, recorded on the TGS and approved; and
- Record discrepancies and non-compliances and make recommendations for rectification.

Table 8-3 provides additional requirements in relation to TMP reviews.

Table 8-3. Additional requirements for TMP reviews

Type of TMP review	Frequency	Considerations
Scheduled	The scheduled TMP review must be undertaken every 6 months or as determined by the relevant G10 Specification or project specific plan (PSP).	The scheduled TMP review must consider the following: • TMP and TGS are approved; • Identify required variations to the TGS, and ensure that they are updated, recorded and approved; • Review any departures or variations to ensure they have been documented and approved; • Speed control effectiveness; and • Construction vehicle entry/egress suitability.
Change generated review	The change generated TMP review must be undertaken when implementing new traffic stages, switches or diversions.	The change generated TMP review must consider the following: • The work site is operating safely; • Delineation is effective with appropriate signage installed for changed conditions; • Safe passage is provided for all road users; • Road Safety Audits are arranged or confirmed as required by <u>Section 8.1.7 TTM road safety audits</u>; and • Accountability for approval and inspection is well understood and documented.
Post-incident or near miss	The post-incident or near miss TMP review must be undertaken following an incident or near miss.	The post incident or near miss TMP review must consider: • Causal factors; • Contributory factors or changes required; and • Identified changes to TGS are completed, approved, recorded and communicated.

Type of TMP review	Frequency	Considerations
		For any incidents or near miss (where required) a safety alert must also be prepared and distributed by the Transport project manager to share learnings with other work sites.

8.1.6 Post completion inspection

A post completion inspection must be undertaken to ensure the site has been demobilised as planned and is safe for opening to traffic. This must be completed by an ITCP qualified person, to ensure:

- The road has been reinstated to the planned condition;
- All temporary speed limits have been removed;
- All signs and devices are removed from the site; and
- No hazards as a result of the work site are present.

If any issues are noted they must be rectified as soon as practical.

An example post completion inspection checklist is provided in [Appendix E – Inspection checklists and tools](#) and may be used to guide this inspection.

8.1.7 TTM road safety audits

A TTM road safety audit (RSA) is a formal technical assessment of a road construction project to identify crash potential and areas of risk that could lead to traffic incidents. The aim of a road safety audit is to undertake an objective assessment of crash risk and identify foreseeable hazards for all road users.

Transport works should develop an audit schedule in line with the relevant divisional quality management framework. The project manager is accountable and must be satisfied that RSAs are completed where required.

The requirement and frequency of an RSA will be specified in the works-relevant G10 or in divisional processes. Additionally, it may be beneficial to perform a pre-opening and finalisation RSA where the temporary or permanent traffic control devices cannot be completed or installed until traffic is switched onto the new alignment.

A TTM RSA must be conducted in accordance with the requirements in the Austroads *Guide to Road Safety Part 6: Managing Road Safety Audits* and must be conducted by an independent and qualified Road Safety Audit team, who must also hold a PWZTMP qualification. TTM RSAs may be completed either internally or externally. RSAs should be carried out during both day and night time to ensure appropriate delineation is in place to suit the conditions.

8.1.8 Client inspections

Additional TTM inspections may be carried out by the client through the Transport Traffic Engineering, Work Health and Safety Branch, Surveillance Officers or other client representatives. Projects should make contact with their client to determine frequency and needed preparations for these inspections.

8.2 Record keeping of TTM documentation

8.2.1 General

Documentation and recording of information related to the TTM is an essential part of the process to ensure that the TTM has been implemented as designed and the controls are achieving the desired outcomes. As required by the State Records Act, records must be retained for seven years as this will also be required in case of an investigation relating to an incident or legal proceedings are commenced.

Records of TTM documentation must be kept and include at a minimum:

- The identification of the job;
- Reference number of the TMP or TGS;
- Date and location of the TTM;
- The name of the person keeping the record; and
- The name of the person responsible for the work.

TTM documentation produced as a result of site-specific works must be kept and include at a minimum:

- Approved TMP developed for the works;
- Completed risk assessments;
- Approved TGSs used, including versions where modifications or updates have been made;
- Other relevant documentation including staging plans;
- Completed inspection checklists that have been undertaken in accordance with Section 8.1 Work site inspections, reviews and audits;
- Records of traffic related incidents that occurred during the works;
- Any departures, and approvals of departures from this Technical Manual; and
- Any other relevant document generated by the process of completing the TTM works.

It is recommended that a folder system be developed to house all relevant TTM documentation for each activity. The format of the folder (e.g. paper based or electronic) and storage requirements will depend on the divisional record keeping requirements.

8.2.2 Generic TGS library

A TGS library is a series of generic TGSs that have been designed by a work group for use at particular locations or for specific work types. The TGS library makes relevant TGSs readily accessible for the person selecting a TGS for specific work activities.

For example, a works depot that is responsible for the regular maintenance of a particular section of the network may collate a series of generic TGSs, from which the nominated PWZTMP or ITCP qualified person can select.

TGSs included in a TGS library must be approved by PWZTMP qualified person. When developing a TGS library, a TGS selection procedure must also be developed in accordance with Section 7.5.2 TGS Selection Procedure, to aid the effective selection of a generic TGS for use.

A process must be developed to ensure the TGSs in the library are:

- Stored appropriately;
- Easily identified and accessed;
- Secure, so that a generic TGS is not changed or removed without authority; and
- Identified as being an approved generic TGS.

A site-specific TGS may become part of the generic TGS library if it is suitable for ongoing use, provided it is approved by a PWZTMP qualified person and endorsed for inclusion in the library.

8.2.3 TTM inspection records

All TTM inspections must be completed in accordance with relevant requirements Section 8.1 Work site inspections, reviews and audits. A record of each TTM inspection must be kept and include the following:

- Reference to the TGS that was used and inspected;
- The start and finish times and location/s of the works;
- The date and time of the inspection;
- Any deficiencies identified and corrective action taken;
- Detail of adjustments and modifications made and by whom;
- Name of person authorising any modifications;
- Any near misses;
- Time of next shift TTM inspection;
- Name of the ITCP qualified person who implemented the TGS;
- Weather conditions and
- Details of on-site traffic controllers;

Daily records should be detailed enough to identify the site, signs and devices which were in place at a given time on a given day. This enables an external party (for example, a coronial request), to establish the conditions and TTM which existed at a specific point in time. Photo and/or video evidence should also be collected and stored in support of other forms of documentation such as checklists, TTM diaries and site registers.

All records of inspections must be made available to Transport personnel, or those working on behalf of Transport.

8.3 Reporting work site incidents

8.3.1 General

For all traffic related incidents, either witnessed or reported, involving the public or from which legal proceedings might arise, the following must be recorded and reported to Transport:

- Type, size and location of signs and devices in use at the time of incident;

- Travel path width and road condition;
- Weather conditions;
- Witnesses present;
- Details of any personal injuries; and
- Extent of any vehicle damage and the vehicle details (e.g. registration).

8.3.2 Reporting procedure for Transport employees

The reporting process in [Table 8-4](#) must be followed by Transport staff performing work for Transport. The SRIMS Helpdesk can be contacted in 1300 131 469 or SRIMS mailbox SRIMS@transport.nsw.gov.au for assistance.

Table 8-4. Reporting requirements for Transport for NSW staff

Type incident	Requirements
Incidents involving Transport employees	All workplace incidents and near misses must be reported immediately to the workplace supervisor who is responsible for investigating the incident. The incident and investigation must also be recorded in Transport's incident reporting system, SRIMS.
Incidents involving non-Transport employees on Transport work sites	All workplace incidents that involve a non-Transport employee including contractors, visitors and members of the public must be reported immediately to the Transport workplace supervisor, who is to determine responsibility for investigating the incident. The incident and investigation must also be recorded in the incident reporting system of Transport, SRIMS.

8.3.3 Reporting to SafeWork NSW

By law, employers or occupiers must report work-related incidents to SafeWork NSW or the relevant workers compensation insurers.

An occupier (of premises or workplace) is someone who manages or has responsibility for a workplace or a particular operation at a workplace, even if they are not the employer.

Serious incidents must be notified to SafeWork NSW immediately as an urgent investigation may need to be undertaken. For Transport internal work, the Work Health and Safety Branch is responsible for notifications to SafeWork NSW.

Serious incidents include where:

- There has been a fatality to an employee, contractor, visitor or member of the public;
- There has been a serious injury or illness, such as when a person:
 - Is placed on a life support system;
 - Loses consciousness;
 - Is trapped in machinery or a confined space; or
 - Has serious burns.
- There is an immediate threat to life, such as major damage to machinery or buildings;
- It is highly visible to the public or has the potential for media interest.

Further details on SafeWork NSW's reporting requirements can be found on the [SafeWork NSW Incident Notification](#) website.

After a serious incident has been reported the immediate area around the incident must not be disturbed (adhering to SafeWork NSW requirements), except where exemptions are provided for under the [WHS Act 2011](#) which include to assist any injured persons, to avoid further injuries and issues or where SafeWork has authorised work to continue.

Appendix A – Traffic management planning templates and tools

INFORMATIONAL APPENDIX

A.1 Scope

This Appendix provides a number of templates that may be used by project teams in relation to temporary traffic management planning. While every attempt has been made to align these templates with the requirements of this Technical Manual, the templates are examples only. The use of these templates might not necessarily fulfil all of the relevant requirements of this Technical Manual. And thus, pertaining to each work site, it remains the obligation of project teams to ensure all relevant requirements of this Technical Manual have been met regardless of the content of these example templates.

Additionally, this Appendix also provides a template that may be used by practitioners who wish to propose general departures to the requirements of this document.

A.2 TMP development templates

A.2.1 General

The templates provided in this Section may be used for temporary traffic management planning. The templates comprise:

- Temporary traffic management strategy template;
- Temporary traffic management plan and approval template; and
- Temporary barrier design statement template.

A.2.2 Temporary traffic management strategy template

For the risk associated with temporary traffic management to be effectively managed, it is important that the conditions and constraints associated with the works are understood. For this to be achieved, the client must collect the relevant data and information and transfer this to the delivery partner to ensure an informed TMP is developed.

MANDATORY INFORMATION

An electronic version of this template is provided [here](#).

Prepared by			
Name:		Role:	
Division:			
Signature:		Date:	
Persons consulted in the development of the traffic management strategy			
Name:		Role:	
Division:		Signature	
Name:		Role:	
Division:		Signature	
Name:		Role:	
Division:		Signature	

Site related information	
Project	
Activity / work	

Site related information

Location				
Current Project Phase	☐ Strategic design	☐ Concept Design	☐ Detailed Design	☐ Other _____

Site related data

Attach a cross section and photo / aerial of location of works

Cross section details	<i>Describe unique features of the cross section</i>		
Setting of works	☐ Urban	☐ Rural	
Existing speed limit/s	<i>(provide examples:</i> <i>Road name 1: limit</i> <i>Road name 2: limit)</i>		
Traffic volumes	Traffic volumes (ADT): _____	Traffic volumes (AADT): _____	
	Peak times AM: _____	Peak times PM: _____	
Traffic composition	☐ OSOM	☐ Heavy vehicles _____ (%)	☐ Permit vehicle routes
<i>If yes provide details</i>	Details		
Details of crash history at location	<i>Provide details of crash history and comment on any patterns of crashes</i>		

Site related data			
Are intersections impacted by the project length?	☐ Yes ☐ No		
<i>If yes provides details</i>	Details		
Vulnerable road users and other facilities	☐ On-street parking	☐ Transport facilities (bus stops)	☐ Clearways
	☐ Cycle paths	☐ Pedestrian paths	☐ Other
<i>If yes provides details</i>	Details		

Constraints			
Significant traffic generators	☐ Community facilities	☐ Events	☐ Schools
<i>If selected provide details</i>	☐ Mines	☐ Other:	
	Location		
	Duration / time restrictions:		
	Impacts:		
Road environment constraints	<i>Heritage, utilities, cuttings, significant cut/fills, bridges, guardrails, limited shoulders, environmental constraints (threatened species)</i>		

Other considerations or comments

Other considerations

Provide additional relevant information not included in the above

OPTIONAL INFORMATION

An electronic version of this template is provided [here](#).

Options assessment			
Option 1:			
☐ Around <i>An around should be considered as the first option.</i>		Description: <i>Provide detail of around strategy, if none available this should also be detailed</i>	
Is this option feasible?		☐ Yes ☐ No <i>If no, provide justification here</i> <i>If yes, complete information below to the best of your ability</i>	
Benefits		Constraints	Estimated option cost
<i>Include potential cost/time benefits</i>			
Duration of works		Hours of operation	Other considerations
	<i>Days</i>	<i>Day / night Or hours (if known)</i>	<i>Including provision for works traffic etc.</i>
	<i>Weeks</i>		
	<i>Months</i>		
Consultation required			Approvals required

Options assessment			
Option 2:			
☐ Around ☐ Past		Description:	<i>Provide detail of around strategy, if none available this should also be detailed</i>
Is this option feasible?		☐ Yes ☐ No	<i>If no, provide justification here</i> <i>If yes, complete information below to the best of your ability</i>
Benefits		Constraints	Estimated option cost
<i>Include potential cost/time benefits</i>			
Duration of works		Hours of operation	Other considerations
	<i>Days</i>	<i>Day / night Or hours (if known)</i>	<i>Including provision for works traffic etc.</i>
	<i>Weeks</i>		
	<i>Months</i>		
Consultation required			Approvals required

Options assessment			
Option 3:			
☐ Around	☐ Past	Description:	<i>Provide detail of around strategy, if none available this should also be detailed</i>
Is this option feasible?		☐ Yes ☐ No	<i>If no, provide justification here</i> <i>If yes, complete information below to the best of your ability</i>
Benefits		Constraints	Estimated option cost
<i>Include potential cost/time benefits</i>			
Duration of works		Hours of operation	Other considerations
	Days	Day / night Or hours (if known)	<i>Including provision for works traffic etc.</i>
	Weeks		
	Months		
Consultation required			Approvals required

If additional options are required use Options 2 and 3 as a template for completion

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Recommendation	
Recommended option	
Justification	
Critical risks and considerations	
Additional comments	

A.2.3 Traffic management plan and approval template

Prepared by			
Name:		Role:	
Card number:		Organisation:	
Signature:		Date:	

Location of works	
Project	
Activity / work	
Location	
Dates relevant for TMP work	DD/MM/YY – DD/MM/YY

Traffic Management Strategy (TMS) Verification	
Has the TMS been received and attached to this TMP?	☐ Yes ☐ No <i>If “no” has been selected a TMP should not be developed until TMS information is obtained</i>
Provide updated information regarding TMS if required	
Current existing speed limit/s	<i>(Road name 1: limit</i> <i>Road name 2: limit)</i>

Traffic Management Strategy (TMS) Verification			
Updated traffic data	Traffic volumes (ADT):		Traffic volumes (AADT): if available
	Hourly traffic volumes :		Operating speed
	Peak times AM:		Peak times PM:
Traffic composition	☐ OSOM	☐ Heavy vehicles _____ (%)	☐ Permit vehicle routes
If yes provide details	Details:		
Site and work specific considerations	<i>Additional to TMS, additional time, with environment or community concerns</i>		
Additional options available	<i>For additional options identified, the process of assessment outlined in the TMS must be completed</i>		

Decision point: Temporary Traffic Management Method	
Was an options assessment completed by the client?	☐ Yes ☐ No If yes, review and provide summary below If no, undertake an assessment
Summary of TMS options	
TTM method	☐ Around ☐ Past ☐ Through Option Selected: <i>nominate option selected from TMS</i>

Decision point: Temporary Traffic Management Method

Justification

Traffic Management Planning

TTM type

☐ Mobile

☐ Low impact

☐ Static

Will lane or shoulder widths need to be modified?

☐ Yes

☐ No

If yes provide justifications and drawings:

Specific road users impacted

☐ Pedestrians

☐ Cyclists

☐ Motorcyclist

☐ OSOM

☐ Freight Industry

☐ Persons with disability, prams or children

☐ Public transport e.g. bus, tram.

☐ Other

If one or more groups selected provide details of impacts and considerations:

Additional location specific requirements to be considered?

Risk assessment

Undertake and attach to this TMP a risk assessment of the proposed works with the determined strategy.

List of sources of information used in risk assessment

Has the risk assessment considered?

☐ Proximity of traffic	☐ Queued traffic	☐ High traffic volume	☐ Traffic speed and compliance behaviour
☐ Traffic composition	☐ Exposure and proximity of workers to live traffic	☐ Length of delays for road users	☐ Traffic generating land use (hospital, mine, school)
☐ Non-compliance with temporary speed limits	☐ Reduced lane and shoulder widths	☐ Compromised access points	☐ Site vehicle access and egress points
☐ Horizontal (curves) and vertical (crests/sags) alignment	☐ Utilities including above and below services	☐ Crash history	☐ Topographical constraints
☐ Sight distances	☐ Emergency services	☐ Car parking impacted	☐ Transport services (bus stops etc)
☐ Access to private and commercial properties	☐ Local road access	☐ Special events or high risk venues	☐ Other _____

Key risks identified as a result of works:

Risk assessment			
Specific controls required:			
Protection of workers	☐ Barriers	☐ Delineation	☐ Other
	Provide details:		
Will a speed restriction be required?	☐ Yes		☐ No
	If yes provide justifications and drawings:		
End queue management strategy:	Provide details of: Calculated end-of-queue length Control required Sight distances		
Delineation of site	Detail how site must be delineated: e.g. reflectivity, non-contradictory signs, devices and delineation		
Emergency service access and notification	Provide details of emergency service strategy for site and who has been contacted		

Relevant Documentation		
Have the following mandatory documents been provided as part of the overall TMP?		
☐ All approved TGS required	☐ Road Occupancy Licence	☐ Plans showing access to local properties or side roads
☐ WHS documentation	☐ Approved list of TTM personnel and contacts	☐ Vehicle movement plans
☐ Traffic incident plans		
STOP: If one of the above documents has not been selected the TTMP cannot be approved		

Relevant Documentation

Other documents provided

☐ Traffic staging arrangements including Traffic Staging Plans	☐ Speed Zone Authorisation
☐ Design drawings	☐ Council permits
☐ Pedestrians and cyclists movement plans	☐ Consultation with public transport operator
☐ Other:	

Monitoring activities required

Person responsible for monitoring *daily* TTM work activities

Name:		Role:	
Unit:		Division:	
Qualification:		Card Number:	
Comments:			

Person responsible for TTM works

Name:		Role:	
Unit:		Division:	
Qualification:		Card Number:	
Comments:			

Review activities required			
Activity	Required		Frequency or details
Shift inspections	☐ Yes	☐ No	
Weekly Inspections	☐ Yes	☐ No	
TMP review	☐ Yes	☐ No	
Road safety audit	☐ Yes	☐ No	
Other:	☐ Yes	☐ No	
Other:	☐ Yes	☐ No	
Comments:			

Endorsed by (when a Principal Contractor undertaking the work)			
Name:			
Role:		Organisation	
Signature:		Date:	

Approval			
<i>I have reviewed the relevant documents for the works and approve works to be completed in accordance with the TTM Plan.</i>			
Name:			
Qualification:		Card Number:	
Unit:		Division:	
Signature:		Date:	

A.2.4 Temporary barrier design statement

Temporary safety barrier design statement template			
TMP Reference #		TMD Reference #	
Site Details	Location	Direction	Barrier Placement
Risk/risk mitigation	Description of the risk(s)		
Barrier systems <i>Hardware must be compliant with the Transport Specification R132</i>	Barrier Type(s)		Terminal(s)
	Transitions (including overlap of different systems)		
	Specific relevant considerations/restrictions		

Temporary safety barrier design statement template

Design requirements – multiple systems will require multiple copies of this table							
System 1		System Deflection* (m)	Minimum Length (m)	Length of Need (LoN) (m)	MASH* Test Level (TL) (1-4)	Minimum Curve Radius	Clear Zone
System name	Required Value <i>From manufacturer's information / R132</i>						
	Designed Value <i>What has been designed</i>						
System 2		System Deflection* (m)	Minimum Length (m)	Length of Need (LoN) (m)	MASH* Test Level (TL) (1-4)	Minimum Curve Radius	Clear Zone
System name	Required Value <i>From manufacturer's information / M23</i>						
	Designed Value <i>What has been designed</i>						
<i>*Where applicable – MASH test results must be used. If NCHRP-350 data is utilised (not yet approved under MASH) this must be stated. All devices deployed after 31st December 2020 must be MASH compliant.</i>							

Temporary safety barrier design statement template

Design Variations					
Equipment and Hardware required <i>List the required hardware including terminal and barrier components</i>	Hardware/Eqpt	Type/notes			Qty
	Delineation (Barrier System)				
	Delineation (End Treatment)				
	Water (litres)				
Installation programme	Sequence				
	Installation/Removal				
Installation Designer Declaration	This design has been prepared in accordance with Transport QA Specification R132 Safety Barrier Systems and associated technical memoranda and standard drawings.				
Installation Designer					
	<i>Name</i>	<i>Date</i>	<i>Signature</i>	<i>Safety Barrier Qualification</i>	<i>Date of Qual</i>
Installed/ Inspected by					
	<i>Name</i>	<i>Date</i>	<i>Signature</i>	<i>Safety Barrier Qualification</i>	<i>Date of Qual</i>

Temporary safety barrier design statement template

Attachments	
-------------	--

A.3 General departures template

Location of works	
Project	
Activity / work	
Location	
Dates relevant for TMP work	<i>DD/MM/YY – DD/MM/YY</i>

Step 1: Details of Departure and risk assessment			
Details of departure			
TCAWS requirement clause reference		Requirement summary	
Location requirement not suitable			
Why is the requirement not suitable?			
Proposed variation to the requirement			
Proposed mitigation measure/s			
Risks introduced as a result of the proposed variation			
Alternative options investigated			

Step 1: Details of Departure and risk assessment			
Supporting Information such as drawings, correspondence etc.	Describe and include attachments if required		
Risk assessment (to be attached to this form)			
New standard being applied:	☐ Alternate Standard / national Practice ☐ Risk Based determination		
Standard applied		Details	
Residual risk rating			
Additional controls required			

Appendix B – Device use requirements

MANADATORY APPENDIX

B.1 Scope

This Appendix provides requirements, recommendations and information on the use, installation, operation and removal of some of the devices in Section 6 Signs and devices.

B.2 Usage procedure: Type 2 (automatic) portable traffic signal systems

B.2.1 Introduction

This Section supplements the requirements of Section 6.6.3 Type 2 (automatic) portable traffic signal (PTS) systems and provides basic information to effectively use portable traffic signals (PTS) to control traffic that have ‘automatic’ functionality. It applies equally to both Transport and other bodies working on roads, such as councils, contractors and public utility authorities. It gives a description of the operational features of the equipment including details for the selection of appropriate signal timings and model situation diagrams showing the required signposting and site layout for the signals.

Type 2 PTS may be used for traffic control applications lasting up to three months. For sites where work will continue for longer periods, without the location of the work site changing, a risk assessment and feasibility analysis must be carried out to determine whether the extended use of portable signals is acceptable or whether a temporary signal installation should be provided. This risk assessment and feasibility analysis must be submitted to Traffic Engineering for review and the concurrence of the Director, Roads and Traffic Engineering obtained for the use of portable signals for longer than three months.

B.2.2 Specification and type approval

PTS systems with automatic functionality must be compliant to Transport Specification TSI-SP-049, Portable Traffic Signal Systems.

Testing of equipment is to be undertaken by the Transport’s Intelligent Transport Systems Branch. After type approval has been issued, the manufacturer or selling agent must affix, to the equipment, a durable marking plate in the following format (Figure B-1):

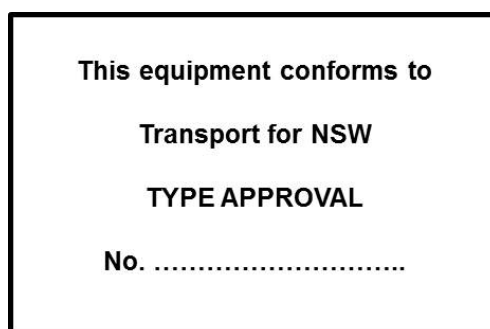


Figure B-1. Marking plate

The number shown on the marking plate must match the number on the type approval certificate issued by the Transport’s Intelligent Transport Systems Branch.

Arrangement for copies of the TSI-SP-049 and type approval testing should be directed to:

- “Transport Traffic Equipment and Standards” at ITSHelpdesk@transport.nsw.gov.au;
- A list of current type approved equipment can be found in Transport specification TS200 (Register of ITS Field Equipment).

Type approved equipment is to be operated in accordance with this Section and the manufacturer's instructions. A record must be kept of the approval and the period of operation of the traffic signals and may be required in court, in case of an accident or traffic infringement, see Table B-1.

Table B-1. Portable traffic signals, record of approval and use

Portable Traffic Signals				
Record of Approval and Use				
Approval: (To be completed by a person with delegated authority)				
Approval is given to the use of portable traffic signals which have been separately type approved to Transport specification, TSI-SP-049 as detailed hereunder.				
Owner:			Signed:	
User:			Title:	
Project:			Date:	
Use: (To be completed by the Users representative)				
Job location:				
Portable signals in service				
Date	Time (24 hour)	Supervisors name, please print	Initials	Mode* of operation
This completed schedule should be kept by the relevant approving authority for a period of two years as a record of the display.				
*Note MAN/1 – Manual Shuttle Operation, MAN/2 – Manual Two-way Operation, VA – Vehicle Activated Shuttle Operation, FT – Fixed Time Shuttle Operation				

Temporary traffic signal installations using fixed equipment and cables are not covered by this Section and will need to be authorised and inspected in the same manner as permanent installations.

B.2.3 General description of system

Each set of PTS equipment with automatic functionality will normally comprise:

- Two signal stands, incorporating signal lanterns, vehicle detectors and control equipment;
- Power pack of batteries or generator;
- Operational spares i.e. chassis and plug panel sub-assembly, controller module, lamp switch module, radio module, signal lantern assembly, lamps and fuses, target board assembly and microwave detector;
- Detector checking unit; and
- Remote manual control box.

Shuttle flow

For shuttle flow on a two-lane, two-way road, one set of PTS is required.

The normal mode of operation is vehicle-actuated using microwave detectors mounted above and integrally with the vehicle signal lanterns. It is also possible to use the signal equipment in either manual control mode (shuttle operation) or fixed-time cycle mode, without the use of the detectors. Under shuttle operation, the operator determines the direction of traffic flows at any time. One direction faces a green display while the other faces a red display.

Plant crossing control

For heavy machinery crossing applications, one set of portable signals is required. This set is used to control traffic on the road with one set of lanterns on each approach. Traffic on the haul road is not usually directly controlled.

Manual operation is safer and more effective in preventing delays to both road traffic and haul vehicles. Under manual control i.e. two-way operation, the operator determines when the road traffic needs to stop to allow the haul vehicles to cross. Both directions face the same display, either green or red.

If aspects are required to be displayed to haul traffic then a second set of signals is necessary which are linked to the road signals. These signals will be arranged to display red when the other set is green and vice versa.

B.2.4 Signposting and traffic arrangements

General

For shuttle flow, the signal stands should normally be located on the shoulder at the start of the taper or at least 30 m clear of the full lane closure. They should be in clear view of approaching drivers. However, if it is found that vehicles in the non-barricaded approach lane are disregarding the signals or are travelling too fast through the work site, then consideration may be given to installing a chicane arrangement in this lane in order to slow approaching traffic as well as allowing conspicuous positioning of the signal lanterns. This should be documented in the site-specific TMP and risk assessment. The dimensions should be selected to suit site conditions, such as the prevailing road geometry, sight distance and vehicle speed. However, it should be noted that the use of the chicane arrangement substantially increases the length of the controlled area and thus imposes the penalty of longer all-red clearance times and increased traffic delays.

Sight distance

Sight distance on the approach to portable or temporary traffic signals must be a minimum of 150 m.

In open road areas or higher speed environments where traffic signals might not be expected, a Traffic Lights symbolic (W3-3) sign with an ___ m (W8-5) sign to indicate distance, should be provided to give advance warning. The position of these signs should take into account the expected queue length from

traffic signals and the stopping distance required for heavy vehicles. Consideration should also be given to installing a portable VMS on each approach as an additional form of advance warning.

B.2.5 Equipment installation

Signal lanterns

Normally one signal lantern is used on each approach thereby requiring only one set of equipment for effective working on one road. PTS with automatic functionality complying with Transport Specification TSI-SP-049 have a facility to connect a second lantern assembly. This would be useful on a wide carriageway.

The lantern must be placed so as to give approaching drivers a conspicuous signal and also to be clearly visible to drivers stopped behind the stop line. The signal aspects should be aimed towards the vehicular traffic whereby vertical adjustment is by adjusting the lantern on the stand and horizontal adjustment is arranged by positioning the stand itself on the ground. In practice, the aiming of the vehicle detectors located on top of the lanterns is more critical than the lanterns so the main emphasis of the aiming operation should be as detailed in Vehicle detectors below.

Generally, the signal unit should be positioned so as to be as nearly as possible in the driver's line of sight. It is essential to locate signals so the driver has sufficient sight distance to stop on a red display. The minimum sight distance to the signal lantern required for stopping depends on the vehicle type and vehicle speed in the approach to the work area. Table B-2 provides a guide to stopping sight distances on level bituminous or concrete surfaces.

Table B-2. Stopping sight distances for cars and laden trucks on level pavements

Vehicle speed (km/h)	Stopping sight distance (m)	
	Cars	Trucks
50	60	80
60	80	100
70	100	130
80	120	160

Note to Table B-2. Increase the stopping sight distance by 2% for each 1% of downgrade. Decrease the stopping sight distance by 2% for each 1% of upgrade.

Signal stands

The signal stands must be securely erected and anchored to prevent excess movement or interference by vandals.

Controller

The controllers allow for two-phase operation only. Each stand has a control module, one of which must be switched to 'master' operation, the other to 'slave' operation. To provide remote manual operation, a special box is connected by cable to the 'master' unit.

The controller is provided with manual controls (switches) for selecting:

- Power: ON/OFF;

- Master/slave selection;
- Mode selection:
 - Manual (shuttle operation);
 - Manual (two way operation);
 - Fixed time/vehicle actuated.
- Manual advance: for manual selection of phases;
- Yellow time: pre-select 4 seconds or 5 seconds;
- All-red time: pre-select in the range 2 seconds to 100 seconds for fixed time and vehicle actuated operation;
- Maximum green time: pre-select in the range 20 seconds to 150 seconds for fixed time and vehicle actuated operation. The minimum green time is fixed at 15 seconds.

The control equipment provides that in the event of internal failure, a loss in radio communications or low battery voltage, the signals revert to flashing yellow.

Communication

The two signal stands, incorporating the signal lanterns and detectors, are linked either by radio or cable control.

Vehicle detectors

Vehicle detectors are used to relay an electrical impulse to the controller when a vehicle approaches. One detector per approach is required.

Microwave detectors may be located on top of the traffic signal lantern and the beam from the transmitter is reflected back to its receiver by the approaching vehicle. The microwave detector will only detect objects moving towards it. Care must be taken to ensure that unwanted vehicles such as construction vehicles, do not actuate the detector.

Response of the microwave detector will depend on the speed of movement and the size of the target object. In general terms, the unit, when positioned to 'look' at an approach, should detect:

- A motorbike moving within the 20 m to 5 m range from the unit; and
- Other larger motor vehicles moving within the 30 m to 5 m range from the unit.

This is assuming that the detector is mounted at a height of between 2.5 m and 3 m and the object being detected is moving at a speed of greater than 5 km/h.

It must be emphasised that the detector should be positioned in such a way that its beam axis is pointed at the centre of the target area and at a distance of approximately 25 m to 35 m ahead of the unit.

For single lane approaches it is suggested that the unit be aimed more towards the road edge. This will result in an increased 'sensitivity' to the desired vehicular movements, but at the same time decrease the sensitivity to vehicles travelling away on the departure side of the street. Although the departing vehicle will not produce a detection output, it can cause a 'swamping' or masking of a signal being reflected by an approaching vehicle.

In practice, aiming of the microwave detector is achieved by moving the complete stand, adjustment in the horizontal plane, and by tilting the lantern by the adjustment at the top of the stand, adjustment in the vertical plane. A visual indicator built into the rear of the detector case assists in the aiming operation. Vehicles should be able to be detected as indicated by the visual monitor at a distance of 5 m to 30 m ahead of the signal stand.

An illustration of how the detectors work is given in *Figure B-2*.

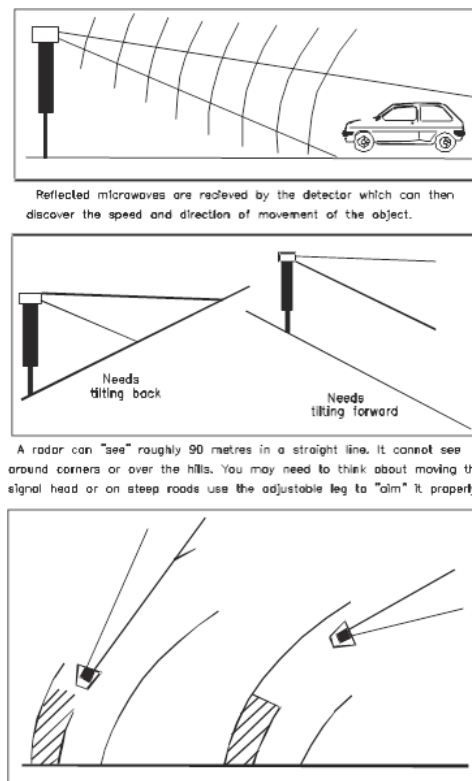


Figure B-2. Illustration of microwave detector operation

B.2.6 Manufacturer's instructions

Operators should familiarise themselves with the manufacturer's Instructions. Field service usually covers:

- Simple fault diagnosis and associated replacement of the faulty modules;
- Routine maintenance including replacement of lamps and other expendable components.

Note: This Section does not replace the field service manual and operating instructions accompanying each set of equipment. It is most important that operators are fully acquainted with the manufacturer's instructions and recommendations before attempting to operate the equipment.

B.2.7 Performance

For manual operation, a remote control box would generally be connected by cable to the 'master' unit, although the equipment can be manually operated at the 'master' stand.

Note: Either unit will operate as a 'master' or 'slave'.

Use of a cable enables the operator to be located safely away from the road in a position where both approaches to the work area are visible. The 'master' unit should then be selected on the basis that the connecting cable should not cross the roadway. For automatic operation, a full-time operator is not required. The operator fixes the initial settings and then monitors the performance intermittently.

Manual mode (shuttle operation)

With manual mode (shuttle operation), the operator controls the sequence in which green aspects are displayed, and also the 'all-red' and 'green' times. The 'minimum green' time cannot be varied by the operator and is fixed at 15 seconds. The 'yellow' time is pre-selected, as appropriate, to 4 or 5 seconds. The 'all-red' time has a minimum value of 2 seconds but the termination of this period is controlled by the operator. The length of the 'all-red' period should be kept to a minimum consistent with the need to clear the controlled area of opposing traffic.

Manual shuttle operation is applicable when any of the following occur:

- A risk assessment has demonstrated that use of PTCO mitigates the risk associated with use of a manual traffic controller;
- Movement of works traffic prevents the effective use of vehicle actuated or fixed time operation; or
- A detector fails, when using the vehicle actuated mode and it is not desired to use fixed time mode.

Manual mode (two-way operation)

With manual mode (two-way operation), the duration and sequence of displays is determined as in Operation (Control modes). The length of the 'all-red' period should be kept to a minimum, consistent with the need to provide for movement of road machinery without interference from normal traffic.

Manual two-way operation is applicable when:

- All traffic needs to be stopped to allow the passage of traffic on a haul road;
- Traffic must be kept out of the work area for an extended period, such as during blasting, priming or full width bitumen sealing.

Vehicle-actuated operation

This mode of operation allows the signals to operate automatically in response to vehicle demands. The signals will change in response to a demand registered by vehicles as they actuate a vehicle detector and the cycle length is adjusted automatically to suit traffic flows.

Vehicle-actuated operation is applicable when either of the following apply:

- Automatic control is required during working hours which allows the signals to operate unattended while still being responsive to changes in traffic flows; or
- Traffic control is required outside working hours so signals can operate unattended provided the power supply is maintained.

The signals must be inspected by the operator at least once per day to ensure that the detectors are functioning correctly and that there are no burnt out lamps and to arrange the daily change of batteries or other servicing. This is particularly important if the site is unattended on weekends. On weekdays, the signals should be checked immediately prior to start and completion of work.

If a detector malfunction is found, the equipment should be switched to fixed-time operation or if an operator is available, to manual operation.

Following the initial switch on and 'master/slave' selection, the operation is as follows:

Step 1: Initially red is displayed on all approaches for a period of at least 10 seconds then each approach in turn receives a green display for its selected 'maximum green' time with a 'yellow' display and 'all-red' display between each green display.

If the signals are switched from the 'manual' or 'fixed-time' modes of operation to the 'vehicle-actuated' mode, the control equipment will automatically register an artificial demand for each phase for the first cycle.

Step 2: After all phases have been called automatically for the first cycle, the signals will change only in response to vehicle demands.

Step 3: If vehicles approach consistently from only one direction, the controller holds the green display on that approach.

Step 4: When a vehicle is detected on another approach, the signals can change in one of two ways:

- When vehicles approach the first phase as a steady stream, the phase holds for the 'maximum green' time before changing to the new approach.

- When the gap between vehicles approaching the first phase is greater than 5 seconds, the signals will change to the new phase, subject to the limitations of the 15 second 'minimum green' time, and after the selected yellow and all-red times.

In the absence of any demand, the signals will revert to 'all-red' until a vehicle is detected. This feature ensures that the signals are then able to give right-of-way to the first approaching vehicle with minimum delay.

If the phase changes at the 'maximum green' time, a new demand is automatically entered for the terminated phase when the phase changes. This ensures that approaching vehicles stopped by the red display will be cleared at an early time. Otherwise, new demands for the terminated phase will only be registered when the arrival of an additional vehicle actuates the detector of that phase.

As a safety feature, when using microwave detectors, an automatic demand will be introduced for any phase or approach which has not received a detector actuation for approximately 200 seconds.

Fixed-time operation

As this form of control does not allow for any response to short-term variations in traffic flow, vehicles can be delayed for no apparent reason when the road is clear. Unattended sites should not be left in fixed time mode (FT).

FT operation is an automatic mode which is not responsive to vehicle demands. The green time is selected by the 'maximum green' switch, and the all-red time is selected by the 'All-Red' switch for each phase. The signals will then cycle in a predetermined order at the times selected. Cycle times can only be varied by manual adjustment of the controller.

FT is most applicable when there is a relatively constant flow of traffic on both approaches. It is also applicable when failure of the vehicle detectors prevents use of the vehicle-actuated mode or when a full time operator is not available.

Limitations on use of microwave detectors

Microwave detectors will only register moving vehicles and so, if for any reason a demand is lost, stationary vehicles waiting at the lights can be ignored by the equipment.

A demand can be lost if vehicles are unable to move off a green display. If vehicles do not start to move within 15 seconds of receiving a green display, the controller will terminate the phase and will ignore the waiting vehicles.

To clear these vehicles, it is necessary to either:

- Wait until a new vehicle joins the queue and actuates the detector;
- Change to manual operation (once traffic is running again, the controller can be returned to vehicle-actuated operation); or
- Wait until the controller (or detector) puts in an artificial demand i.e. it is programmed to place such an artificial demand approximately 200 seconds after the phase was last demanded.

Flashing yellow feature

The control equipment automatically switches all yellow aspects to 'flashing yellow' within 0.5 seconds when any hazardous or incompatible conditions occur in the operation of the equipment, as required by Transport Specification TSI-SP-049.

In the 'flashing yellow' mode, the red and green aspects remain blacked out, and all yellow aspects flash at a rate between 55 and 65 flashes per minute.

If the equipment is allowed to operate for a prolonged period on flashing yellow without a battery change, the signals will eventually turn off.

B.2.8 Operation

Control modes

Although the layout and switch configuration can differ, all controllers have basically the same functions:

- All red – All displays red. Any running sequence is cleared first;
- Manual (MAN) – Control over green displays is via switches or push buttons on the controller. Control can be passed from one phase to another;
- Fixed time (FT) – Control transfers from one phase to the next in a cyclical manner. Each phase receives the green display regardless of traffic flow; and
- Vehicle actuated (VA) – The controller responds to signals from the vehicle detectors. With no demands present the signals will rest in the all-red period.

The mode of operation (manual, vehicle actuated or fixed time) should be selected giving consideration to the operating conditions of the particular site as discussed in more detail in 'Performance'.

Time settings

General

These are shown in Table B-3.

Table B-3. General time setting

Mode	Operation	All red	Minimum green	Maximum green	Yellow
MAN/1	<i>Shuttle</i>	<i>M</i>	<i>F</i>	<i>M</i>	<i>S</i>
MAN/2	<i>Two-way</i>	<i>M</i>	<i>F</i>	<i>M</i>	<i>S</i>
FT	<i>Fixed Time</i>	<i>S</i>	<i>F</i>	<i>S</i>	<i>S</i>
VA	<i>Vehicle Activated</i>	<i>S</i>	<i>F</i>	<i>S</i>	<i>S</i>

Note to Table B-3: The following abbreviations apply:

F – Fixed at 15 seconds

M – Set the manual control switch each cycle

S – Needs to be selected and pre-set by the operator for each site

Yellow time

Estimate approach speed. Select the yellow time from Table B-4

Table B-4. Yellow time setting

Approach speed	Yellow time
less than 75 km/h	4 seconds

All red time

Measure the distance between the stop lines at each traffic signal. Select an appropriate all-red time from Table B-5 or Table B-6 depending on the minimum clearance speed is 20 km/h or 40 km/h respectively.

Maximum green time

Select a maximum green time from Table B-5 or Table B-6 depending on the minimum clearance speed. In FT mode, adjust the maximum green times by allowing 3 seconds for each vehicle queued at the end of the all-red period on each approach.

Note: *The minimum setting is 20 seconds.*

In VA mode, the green time will gap off when traffic clears and only run to the maximum if there is no demand for the other phase. If long queues are regularly occurring in one approach, try increasing the maximum green time on that approach only.

Table B-5. Initial signal time settings (low speed)

Distance between stop lines at traffic signals (m)	All red period* (seconds)	Maximum green period (seconds)
0 to 30	2	30
30 to 45	5	35
45 to 75	10	35
75 to 105	15	40
105 to 135	20	40
135 to 165	25	45
165 to 195	30	45
195 to 250	40	50
250 to 310	50	50
310 to 365	60	60
365 to 415	70	70
415 to 465	80	80
465 to 525	90	90
525 to 575	100	100

Note* to Table B-5: Based on a minimum clearance speed of about 20 km/h.

Table B-6. Initial signal time settings (high speed)

Distance between stop lines at traffic signals (m)	All red period* (seconds)	Maximum green period (seconds)
0 to 50	2	30
50 to 90	5	35

Distance between stop lines at traffic signals (m)	All red period* (seconds)	Maximum green period (seconds)
90 to 150	10	35
150 to 210	15	40
210 to 270	20	40
270 to 330	25	45
330 to 390	30	45
390 to 500	40	50
500 to 620	50	50
620 to 730	60	60
730 to 830	70	70
830 to 930	80	80
930 to 1050	90	90
1050 to 1150	100	100

Note* to Table B-6: Based on a minimum clearance speed of about 40 km/h.

Setting up

The set-up steps given below must be followed and read in conjunction with the manufacturer's requirements:

1. Set up stands with signal aspects facing oncoming traffic
2. Connect controller to the generator and start generator
3. **DO NOT SWITCH CONTROLLER ON**
4. Set red and green times as required
5. Select mode switch
6. Ensure that the shuttle lane or haul road is clear and then switch on controller
7. Controller will serve each phase in turn, clearing the initial demand.

Note: Waving a hand smartly toward the detector should place a call on each side for checking VA mode if selected. This will verify that signals are operating properly in this mode even in the absence of traffic.

Troubleshooting in VA mode

Details are shown in [Table B-7](#).

Table B-7. Troubleshooting in VA mode

Problem	Possible cause	Remedy
Long queues	- Green setting too short - Detector fault - Road capacity exceeded	- Increase setting - Call service. - Call supervisor
Signals do change after one stream has stopped even though traffic is waiting	Detector fault	Call service. Operate signals in manual or FT mode until service arrives
Green period always same length	- Detector fault - Green setting too short - Traffic flow very light - Traffic flow too heavy	- Call service - Increase green time setting - No action - Call supervisor
Traffic still in shuttle lane at start of opposite green	- Traffic running the red light - All-red setting too short	- Call police - Increase all-red setting
Long gap between last vehicle clearing shuttle lane and start of next green	- All-red setting too long - Detector fault	- Decrease all-red setting. - Call service
Signals do not remain on red in absence of traffic	Detector fault	Call service

Examples

Site details

Work area 100 m long in a rural highway situation (see [Figure B-3](#)):

- Approach speeds 80 km/h;
- Distance between stop lines 160 metres; and
- Minimum clearance speed 20 km/h.

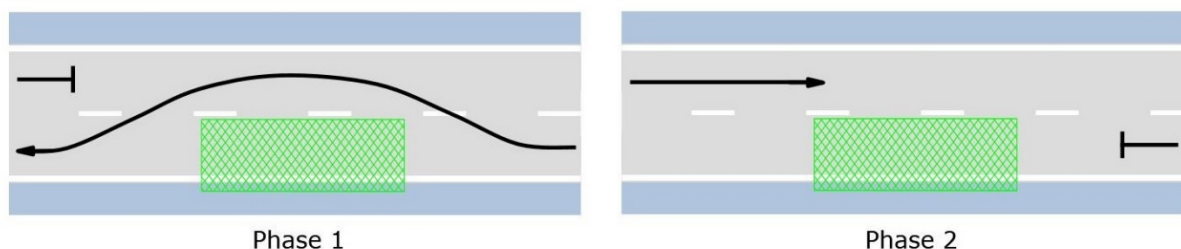


Figure B-3. Illustration of site layout

Example of selecting times for “fixed time” operation

1. From time setting instructions in Maintenance (Transport equipment) and Table B-4 and Table B-5, select initial controller settings:

Table B-8. Initial controller settings

Yellow	5 seconds
All-red	25 seconds
Max green:	
Phase 1	45 seconds
Phase 1	45 seconds

2. Switch on and observe for three cycles:

- If all vehicles clear the work area during the all-red period do not adjust “all-red” setting. If vehicles do not clear the work area extend the “all-red” period;
- Count the number of vehicles queued at the end of the all-red period on each approach. Assume the following vehicles queued at the end of the red period:

Table B-9. Number of vehicles queued

	Phase 1	Phase 2
First cycle	10	1
Second cycle	15	6
Third cycle	9	4
Max queued vehicles for three cycles	15	6

3. Adjust “maximum green” times for the two phases:

- Phase 1 – $15 \times 3 = 45$ seconds;
- Phase 2 – $6 \times 3 = 18$ seconds (use the minimum setting of 20 seconds).

4. Check regularly that vehicles are not experiencing unnecessary delays due to incorrect settings.

Example of selecting times for “vehicle-actuated” operation

1. From time setting instructions in Table B-4 and Table B-5, select initial controller settings as for FT operation.
2. Switch on and observe for three cycles:
 - Check all-red setting as for FT operation;
 - Time the green period for each phase using a stop watch. Assume the following green times were measured:

Table B-10. Green times

	Phase 1	Phase 2
First cycle	45	45
Second cycle	45	35
Third cycle	45	30

3. Try extending the maximum green time for phase one by 5 to 10 seconds.
4. If phase one is still running to its maximum green setting, repeat Step 3.
5. Check regularly that vehicles are not experiencing unnecessary delays due to incorrect settings.

B.2.9 Maintenance

General

The equipment should be maintained in good working condition and expendable items, such as traffic signal lamps replaced immediately after they fail. If faults develop, the signals should be taken out of service and alternative traffic control arrangements made. The signals should not be returned to service until the faults have been rectified.

Transport equipment

Maintenance is required on the various items of Transport plant, for example:

- Signal stands, incorporating signal lanterns, vehicle detectors and control equipment;
- Battery packs; or
- Battery charger.

This maintenance is to be addressed under established practice for major plant items. Signal lanterns, vehicle detectors and controllers are to be returned for repair with associated operation and/or maintenance manuals and wiring diagrams to Transport Works Centre as follows:

The Works Supervisor
 Transport Services
 Traffic Workshop
 129A Orchardleigh Street
 Yennora 2161
 Telephone (02) 9794 4747

Equipment problems, such as faulty wiring that can be corrected in the field may be rectified by a qualified electrician or radio service technician. Expendable items, such as traffic signal lamps and fuses should be replaced in the field and small stocks of such items should be kept in field offices.

Note: The signal lamps are a special type with quartz envelopes. These quartz envelopes should never be touched with bare hands as they are easily damaged. The lamps should always be held by the metal bases and fitted using cotton gloves.

B.2.10 Signage

Sight distance on the approach to traffic signals must be 150 m to the primary signal face and the approach speed of traffic must be no higher than 60 km/h. In open road areas where traffic signals might not be expected, a Traffic light symbolic (W3-3) or (T1-30) sign with an ___ m (W8-5) sign to indicate distance, should be provided to give advance warning. The position of these signs should take into account the expected queue length from traffic signals and the stopping distance required for heavy vehicles, with this information included in the TMP and/or risk assessment. The STOP HERE ON RED SIGNAL (R6-6) sign must be located 6 m in advance of the portable traffic signal and consideration given to the application of a solid white line (200-300 mm in width) adjacent to the R6-6 extending to the centre of the road. Consideration should also be given to installing a portable VMS on each approach as an additional form of advance warning. An example of a work site layout with PTS is provided in [Figure B-4](#).

Additional instructions on the associated signs and use of portable traffic signals are given in [Section 6 Signs and devices](#).

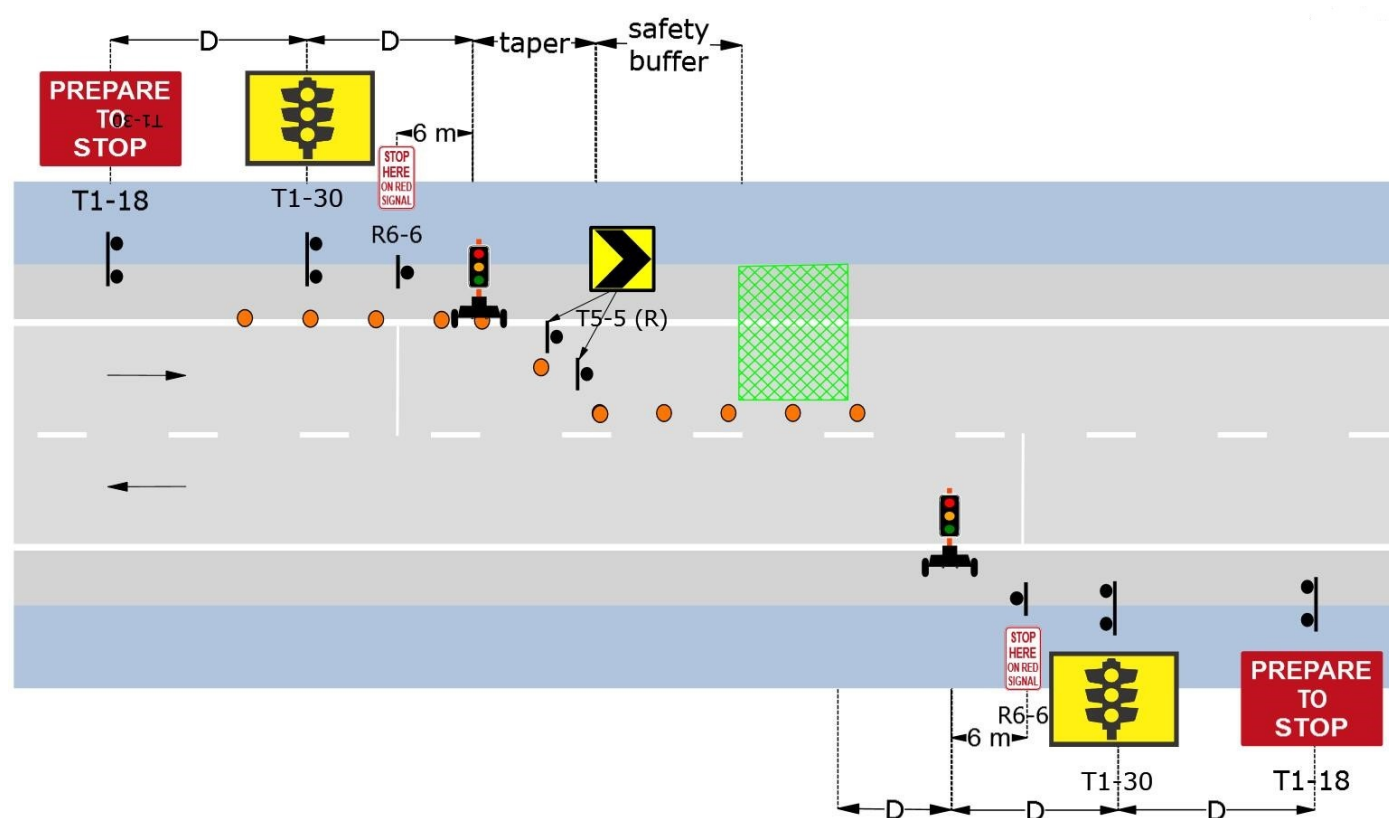


Figure B-4. Example work site layout with PTS

B.3 Usage procedure: Type 1 (manual) portable traffic signal systems

B.3.1 Purpose

This Section supplements the requirements of [Section 6.6.4 Type 1 \(manual\) portable traffic signal \(PTS\) systems](#) and provides the operational requirements for Type 1 (manual) PTS in NSW with or without the use of a boom arm. For the purposes of this manual:

- Type 1 (manual) PTS devices must be approved in accordance with [Transport Specification No. TSI-SP-059 Type 1 Portable Traffic Signals](#).

- Type 1 (manual) PTS devices with a boom arm (Type 1 PTS-B) must be approved in accordance with [Transport Specification No. TSI-SP-081 Type 1 Portable Traffic Signals with Boom Barrier](#).

Reference to Type 1 PTS in this Section includes Type 1 (manual) PTS and Type 1 (manual) PTS-B. This Section must be read in conjunction with the Transport Specification TSI-SP-059 Type 1 Portable Traffic Signals.

B.3.2 Installation and operation

Installation and removal

When installed, Type 1 PTS must be:

- Located where there is a sight distance on the approach to the Type 1 PTS of at least 150m;

Note: See [Table B-2](#) which provides a guide to stopping sight distances on level bituminous or concrete surfaces.

- Located in the line of sight of approaching road users, such that all lanterns are clearly visible and unobscured at all times;
- Installed such that they can be securely erected and anchored to prevent excess movement for all likely weather conditions;
- Located at a safe distance from the vehicle travel path, so that loads or turning vehicles will not impact the unit while remaining within the drivers' line of sight;
- Located such that its visibility is not diminished due to the proximity of other warning devices (such as flashing lights) or other plant and vehicles;
- Installed so that the top of the lantern, when erected, is between 2.5 m and 4 m from the surface of the road at the foot of the stand;
- Installed with target boards and visors fitted and in place during operation;
- Installed in accordance with the manufacturer's operation manual and guidelines; and
- Secured so as to not be at risk of being displaced as a result of wind or other disturbance.

When installing a Type 1 PTS, appropriate manual handling techniques should be used in the set-up, pack up and transportation of the device.

At the end of the work shift the Type 1 PTS must be removed from the traffic lane. The PTS must be stored behind Transport approved safety barriers or as far away from the edge of the traffic lane as is practical in a position determined suitable based on a documented risk assessment and detailed in the TMP. For more information refer to Austroads [Guide to Road Design](#) Part 6: *Roadside Design, Safety and Barriers*.

Operation

When installed, Type 1 PTS must be operated by an authorised traffic controller who is trained and competent in the operation of the product. The system must not be left unattended during operation. The authorised traffic controller must maintain direct control of the Type 1 PTS and must do so in accordance with the operating principles and procedures used by a manual traffic controller with a hand held STOP/SLOW bat in single lane operation.

During the operation of a Type 1 PTS:

- Queue lengths and driver behaviour must be regularly monitored and reviewed;
- If adverse driver behaviour is observed, such as swerving, "drive through on red" or end-of-queue incidents, additional control devices such as cones, temporary kerbing or bollards should be installed. These events and any additional controls implemented must be recorded;

- If adverse driver behaviour continues to be observed, the Type 1 PTS must be replaced with an alternative form of traffic control and recorded;
- If two or more traffic controllers are required to operate the system, each traffic controller must be equipped with radio communication;
- Traffic controllers must not use STOP/SLOW signage that may conflict with the Type 1 PTS signage; and
- Items not directly related to the operation of the Type 1 PTS must not be attached to the unit (such as roadwork signage) unless they are in accordance with the relevant specification or approved by Traffic Engineering.

Low visibility and night time operation

When operating a Type 1 PTS in low visibility environments such as fog, low light or under night conditions, the area in which it is to be used must be sufficiently lit to ensure the lanterns on the Type 1 PTS do not negatively impact the night time vision of drivers.

When operating Type 1 PTS in night conditions, any potential impacts on night time vision and mitigation strategies must be documented in the Traffic Management Plan and associated risk assessments.

The risk of night time vision deterioration can be dependent on the:

- Exposure time experienced (e.g. short period just driving past or extended waiting for ability to proceed);
- Type of work that is being protected;
- Distances between the lights and the works; or
- Use and placement of other artificial lighting systems (e.g. lighting towers).

B.3.3 Traffic controller requirements

General

All requirements for traffic controllers provided in this document must be complied with.

Risk assessment

In relation to traffic controllers operating a Type 1 PTS, a risk assessment must be conducted to determine the number of traffic controllers to operate the Type 1 PTS.

The risk assessment must also be conducted to determine the safe location of the traffic controller/s in relation to their visual contact with the Type 1 PTS and traffic. See additional requirements below for the location of traffic controllers operating a Type 1 PTS.

Note: One traffic controller may operate up to two Type 1 PTS devices.

Location of traffic controllers

Traffic Controllers operating the Type 1 PTS must:

- Be located off the road pavement and out of the travel lane or path of vehicles;
- Have a clear and predetermined escape route;
- Maintain clear visibility of the Type 1 PTS and the front of traffic queues;
- Maintain clear visibility of both Type 1 PTSs and the front of traffic queues for each approach when one traffic controller is operating two barriers; and
- Be located behind a safety barrier where available.

The risk assessment which determines the location of the traffic controller/s must also consider:

- Weather conditions, lighting, road geometry and roadside objects that may obstruct visibility or the escape route of the traffic controller/s;
- Roadside conditions such as long grass or vegetation, uneven and sloping surfaces or other hazards that may be a risk to workers on foot;
- Radio signal range between the PTS remote controls and the traffic signal lanterns; and
- Visual contact of traffic controllers with motorists to encourage compliance.

B.3.4 Signage

Advanced warning must be placed to alert motorists of the Type 1 PTS ahead. Details of the associated signs and use of portable traffic signals are given in [Section 6 Signs and devices](#). When the device is not in operation, these signs must be covered or removed. A work type example for the use of a PTCD is provided in [Figure B-4](#).

B.3.5 Critical faults

If a critical fault occurs, the Type 1 PTS revert to an all-red display on all lanterns within five seconds. When this occurs traffic controllers and STOP/SLOW bats must be available to perform traffic control for each Type 1 PTS. The traffic management plan must specify the management of equipment and traffic control failures to ensure the ongoing safety of workers and road users.

B.4 Usage procedure: portable boom barriers

This Section supplements the requirements in [Section 6.6.5 Portable boom barriers](#) and outlines the operational requirements for portable boom barriers in NSW. At the time of publication, there is no Transport specification for portable boom barriers.

B.4.1 Installation and operation

Installation and removal

When installed, portable boom barriers must be:

- Located where the sight distance between the device and oncoming traffic is at least 1.5D;

Note: A distance of less than 1.5D may be used if a site-specific risk assessment has been undertaken, and documented in the TMP, and the departure is approved in accordance with [Section 2.8 Departures from this Technical Manual](#), with additional measures adopted to ensure traffic controller safety.

- Located clear of overhead hazards or obstructions;
- Located such that the end of the boom is 500 mm from the edge of the adjoining travel lane;
- Installed such that additional weight can be added to the stability legs via sandbags or other stabilisation methods, if winds greater 50 km/h are expected during operation;
- Installed when there are sufficient gaps in traffic to allow for the predictable and safe lowering of the boom without adversely affecting driver behaviour; and
- Installed in accordance with the manufacturer's operation manual and guidelines.

When installing a portable boom barrier, appropriate manual handling techniques should be used in the set-up, pack up and transportation of the device.

At the end of the work shift, the portable boom barrier must be removed from the traffic lane. The barrier must be stored in a safe location behind Transport approved safety barriers or as far away from the edge of

the traffic lane as is practical in a position determined suitable based on a documented risk assessment and detailed in the TMP. For more information refer to the Austroads [Guide to Road Design](#) Part 6: *Roadside Design, Safety and Barriers*.

Operation

When installed, portable boom barriers must be operated by an authorised traffic controller who is trained and competent in the operation of the specific proprietary product. The system must not be left unattended during operation. The authorised traffic controller must maintain direct control of the portable boom barrier and must do so in accordance with the operating principles and procedures used by a manual traffic controller with a hand held STOP/SLOW bat in single lane operation.

During the operation of a portable boom barrier:

- Queue lengths and driver behaviour must be regularly monitored and reviewed;
- If adverse driver behaviour is observed, such as swerving, adverse braking or end-of-queue incidents, additional control devices such as cones, temporary kerbing or bollards should be installed. These events and any additional controls implemented must be recorded;
- If adverse driver behaviour continues to be observed, the portable boom barrier must be replaced with an alternative form of traffic control and recorded;
- If two or more traffic controllers are required to operate the system, each traffic controller must be equipped with radio communication;
- Traffic controllers must not use STOP/SLOW signage that may conflict with the portable boom barrier signage; and
- Items not directly related to the operation of the portable boom barrier must not be attached to the unit (such as roadwork signage) unless it has been approved by Traffic Engineering.

Low visibility and night time operation

When operating a portable boom barrier in low visibility environments such as fog, low light or under night conditions, the area in which it is to be used must be sufficiently lit to ensure the visibility of the boom to road users.

B.4.2 Traffic controller requirements

General

All requirements for traffic controllers provided in this document must be complied with.

Risk assessment

In relation to traffic controllers operating portable boom barriers, a risk assessment must be conducted to determine the number of traffic controllers to operate the portable boom barriers.

The risk assessment must also be conducted to determine the safe location of the traffic controller/s in relation to their visual contact with the portable boom barriers and traffic. See additional requirements below for the location of traffic controllers operating a portable boom barrier.

Note: One traffic controller may operate up to two portable boom barrier devices.

Location of traffic controllers

Traffic controller/s operating the portable boom barriers, must:

- Be located off the road pavement and out of the travel lane or path of vehicles;
- Have a clear and predetermined escape route;

- Maintain clear visibility of the portable boom barrier and the front of traffic queues;
- Maintain clear visibility of both barriers and the front of traffic queues for each approach when one traffic controller is operating two barriers; and
- Be located behind a safety barrier where available.

The risk assessment which determines the location of the traffic controller/s must also consider:

- Weather conditions, lighting, road geometry and roadside objects that may obstruct visual contact or the escape route of the traffic controller/s;
- Roadside conditions such as long grass or vegetation, uneven and sloping surfaces or other hazards that may be a risk to workers on foot;
- Radio signal range between the portable boom barrier and the remote controls; and
- Visual contact of traffic controllers with motorists to encourage compliance.

B.4.3 Signage

Advanced warning must be placed to alert motorists of the portable boom barrier ahead. The placing of the sign T1-272n (*Figure B-5* below) must be in accordance with this document. When the device is not in operation, these signs must be covered or removed.



Figure B-5. Portable boom barrier ahead sign (T1-272n)

The STOP sign (R1-1) must be fixed to the boom barrier (see *Figure B-6*), and positioned so that it is vertically and horizontally in the centre of the boom, and is clearly visible to approaching road users.



Figure B-6. Stop sign (R1-1)

Additional instructions on the associated signs and use of portable boom barriers are given in [Section 6 Signs and devices](#).

A work type example for the use of a PTC is provided in [D.4.4 Static: Dual lane closure for work in fast lane - 3 lane / 2 way \(formerly TCP 90\)](#).

B.4.4 Critical faults

If a critical fault occurs, traffic controllers with STOP/SLOW bats must be available to perform traffic control for each portable boom barrier.

The traffic management plan must specify the management of equipment and traffic control failures to ensure the ongoing safety of workers and road users.

B.5 Usage procedure: illuminated flashing arrow signs

B.5.1 Introduction

This Section supplements the requirements in Section 6.9.2 Illuminated flashing arrow signs and provides the basic information to effectively use illuminated flashing arrow signs. These signs comprise a matrix of lamps or LED aspects in the form of an arrow that is flashed in a cyclic manner to either the left or right, indicating the direction in which approaching vehicles are to pass.

The equipment is to be operated in accordance with this Section and with the manufacturer's instructions. As part of the daily routine tasks and record keeping, a log must be maintained to record the location and period of display of the signs. This documentation may be required in court in case of an accident or other incident, such as a traffic infringement. Other bodies working on roads are encouraged to follow these practices where they are applicable. See Section 8.2 Record keeping of TTM documentation.

Flashing arrow signs are intended to be applied primarily where a lane is closed or a diversion of traffic is required, typically on a multi-lane carriageway. They may also be adapted for mobile plant operation where only part of the road is blocked by the road plant, but a clear direction to traffic is required as to which side of the plant traffic should pass through the dynamic work site. An example of this is longitudinal linemarking.

Note: This Section does not replace the field service manual and operating instructions for each set of equipment. It is important that operators are fully acquainted with the manufacturer's instructions and recommendations before attempting to operate the equipment.

B.5.2 Approvals and specifications

Illuminated flashing arrow signs must comply with the relevant Australian Standards, where they exist, and Transport Specifications TSI-SP-060 Illuminated Flashing Arrow Signs. The relevant Australian Standards are listed in TSI-SP-060. General operating instructions are given in this Section.

Testing of equipment is to be undertaken by the Transport's Intelligent Transport Systems Branch. After type approval has been issued, the manufacturer or selling agent must affix, to the equipment, a durable marking plate in the format provided in Figure B-7:

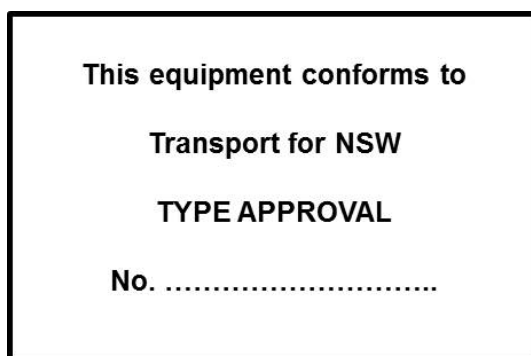


Figure B-7. Marking plate

The number shown on the marking plate must match the number on the type approval certificate issued by the Transport's Intelligent Transport Systems Branch.

Arrangement for copies of the TSI-SP-060 and type approval testing should be directed to:

- “Transport Traffic Equipment and Standards” at ITSHelpdesk@transport.nsw.gov.au;
- A list of current Type Approved equipment can be found in Transport specification TS200 (Register of ITS Field Equipment).

B.5.3 General description of system

Illuminated flashing arrow sign equipment

The following three size designations are used:

- Type A;
- Type B; and
- Type C.

Type A and Type B signs are suitable for mounting permanently on the rear of a road construction vehicle (fixed sign arrangement) or on the cab of a truck or utility vehicle (adjustable sign arrangement). Cab-mounted signs can be rotated to face either the front or the rear of the vehicle. The sign can also be rotated and locked in the face-down position for transport when the sign is not in use. They may be powered from the vehicle's electrical system or from a separate power source.

Type C signs are trailer mounted with integral generator, back-up battery supply and control equipment. Provision is made for the sign to be lowered, rotated and locked for transport or when the sign is not in use.

All signs can operate in one of four modes as detailed in [Table B-11](#):

Table B-11. Modes of operation

Mode	Rear monitoring for Type C	Flashing lamps
Arrow right (AR)	— — — — ■	Shaft and the right-side arrow head
Arrow left (AL)	■ — — — —	Shaft and the left side arrow head
Double arrow (DA)	■ — — — — ■ synchronised flash	Shaft and both the left and the right-side arrow heads
Warning (W)	■ — — — — ■ alternating flash	Pairs of diagonally opposite lamps. Gives a general message of caution or draws attention to an important traffic control sign.

Vehicle-mounted signs

Type A signs are suitable for attachment to light vehicles, such as cars, panel vans or utilities or small plant items. Type B signs are suitable for attachment to heavy vehicles, such as trucks and large plant items.

They are generally used on dynamic works, such as longitudinal linemarking, mobile survey vehicles, patrol vehicles and on shadow trucks protecting groups of workers.

Trailer-mounted signs

Type C signs are self-contained trailer mounted units particularly designed for use on high speed roads, such as rural divided roads where driver expectations are high.

They are generally used at short-term or long-term lane closures for work requiring one or more lanes to be closed for one or more shifts. Where the flashing arrow signs are to operate overnight or over weekends, the associated signs and devices need to be either reflective or lit.

Associated signposting and traffic arrangements

Any additional or associated signposting or traffic arrangements that may be required to be used with flashing arrow signs must conform to the requirements of this Technical Manual.

Using flashing arrow signs tends to downgrade the effectiveness of other devices at the work site so it is essential that the associated signs and devices are in very good condition and special care be taken in their erection. The visibility distance for Type A, B and C flashing arrow signs is 500 m, 1000 m and 1500 m respectively. Associated signs should be located in clear view of approaching drivers, generally on the left side of the road. However, on winding alignment it may also be necessary to erect a sign or signs on the right hand side for clear viewing. Duplicate signs, on the right hand side of the road, may be considered when a driver's view can be obscured by alignment, buildings or heavy traffic. The use of larger signs may be considered for high approach speeds.

B.5.4 Equipment installation

Lamps

The brightness of the lamps is adjusted by an automatic dimming control which dims the light output for night conditions. A photocell detects the ambient light conditions that trigger the dimming facilities.

The flash rate for flashing arrow displays (AL, AR, and DA) can vary from 50 to 60 per cent on-time and 50 to 40 per cent off-time. For diagonal flashing displays (W), the on-time and off-time is the same. The repetition rate for all modes is between 35 and 40 cycles per minute.

Two monitoring lamps are provided at the rear of the Type C sign to allow workers to monitor the mode of operation. These lamps flash as shown in [Table B-11](#).

Sign boards

Generally, the flashing arrow sign unit should be positioned as near as possible in the driver's line of sight. It is important to always locate signs to give the driver maximum visibility and time to understand and react to the sign message. The visibility distance of the sign i.e. the distance at which the motorist can first become aware of the flashing arrow, varies as shown in [Table B-12](#). The minimum sight distance that should be provided to the flashing arrow sign depends both on the vehicle type and vehicle speed in the approach to the work area.

[Table B-12](#) provides a guide to the sight distances appropriate for the three types of sign. On high speed roads, every effort should be made to position the sign to achieve the desirable minimum sight distance:

Table B-12. Typical sight distance for placement of flashing arrow signs

Sign type	Visibility distance (m)	Sight distance	
		Desirable minimum	Absolute minimum
A	500	250	150
B	1000	500	300

Sign type	Visibility distance (m)	Sight distance	
		Desirable minimum	Absolute minimum
C	1500	750	450

Note to Table B-12: Increase the sight distance by 2% for each 1% of downgrade. Decrease the sight distance by 2% for each 1% of upgrade.

The signs must be securely erected and anchored to prevent wind movement or interference by vandals. This is especially true for Type C signs as these are intended to be operated unattended.

The signs must also be aimed carefully to direct their display to the approaching vehicles. It is important that the sign be aimed to vehicles within the 'critical zone' which covers the distances where drivers have time to react to the message and change lanes or stop if necessary. This is particularly important if the approach alignment of the road is not straight.

Sighting is usually done by means of a 'sighting' device which facilitates aiming of the sign display. Such an aiming device must be substantially free of parallax error and must make allowance for the inherent downcast in the sealed-beam lamps. In the absence of a 'sighting' device fitted to the sign, aiming needs to be undertaken by "trial and error".

Table B-13 lists the recommended aiming distances from the sign near the start of the taper defining the lane closure:

Table B-13. Aiming distances for various approach speeds

Approach	Aiming distance (m)		
speed (km/h)	sign type		
	A	B	C
less than 60	60	120	180
60	100	200	300
80	140	280	420
100	200	400	600
greater than 100	260	520	780

Note 1 to Table B-13: The aiming distance should not exceed the sight distance.

Note 2 to Table B-13: The recommended aiming distances for the Type A sign is based on the stopping distance plus the reaction distance for 2.5 seconds of travel PLUS an allowance of 30 metres for siting the sign past the start of the taper

Note 3 to Table B-13: The aiming distances for Type B and C signs are two and three times of those for the Type A sign to take advantage of their greater size and visibility

Note 4 to Table B-13: Type C signs are not usually used on low speed roads.

Figure B-8 is an illustration of aiming and sight distance, showing the relationship between these two distances used when setting up flashing arrow signs.

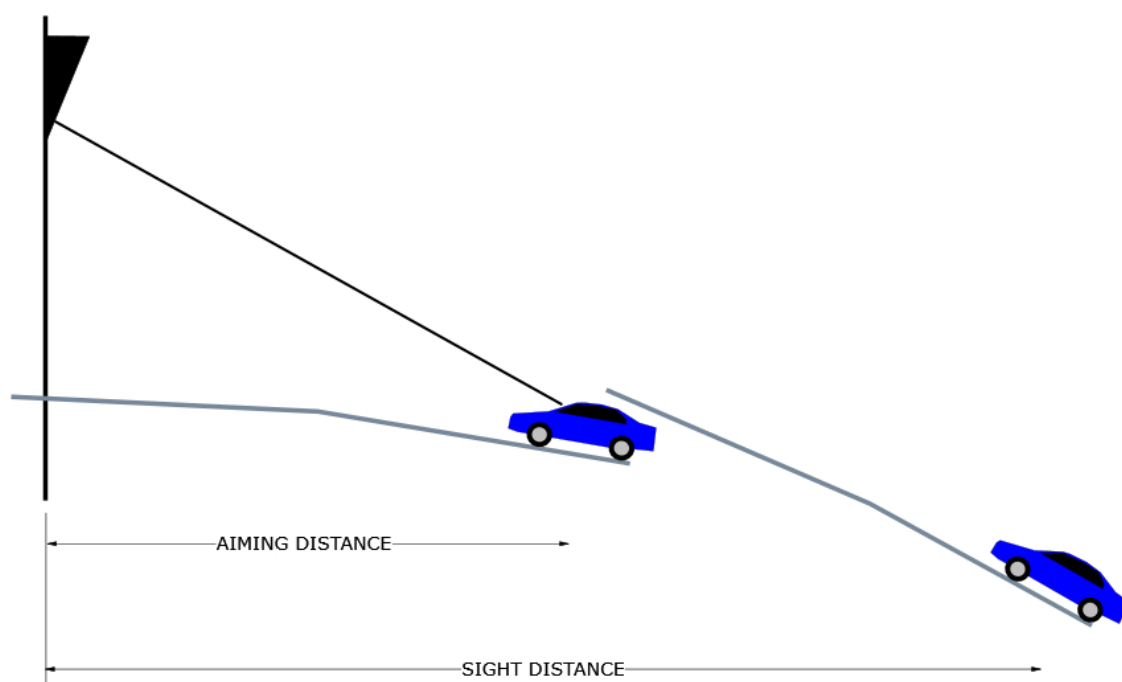


Figure B-8. Illustration of aiming and sight distance

Controls

The controls available are:

- Power: On/Off;
- Mode selection; and
- Mode monitoring (Type C signs only).

B.5.5 Operation

Control modes

Although the layout and switch configuration can differ, all controls have basically the same functions, namely:

- Arrow Right (AR);
- Arrow Left (AL);
- Double Arrow (DA); and
- Warning (W).

Flash rate

The flash rate is fixed to the levels given in the specifications and cannot be varied. It is important to check that all lamps are functioning to give the correct display selected by the controls.

Dimming

The dimming facilities are automatic, but care should be taken that any incident light from street lighting or vehicle headlights falling on the photocell does not adversely interfere with the operation of the dimming feature. The photocell should be occasionally cleaned to maintain brightness.

Setting up

The equipment should be set up to the manufacturer's instructions with particular care to the siting and aiming of the signs to suit the layout arrangement being used and the geometry of the road approach to the sign.

Troubleshooting

The monitoring lamps should be regularly observed to confirm that the sign is operating in the appropriate mode for the work site. It is also necessary to regularly inspect the front panel of the sign to ensure that all lamps are clean and alive.

If one lamp is not operating, check it and if it is blown, replace it with a new lamp. If all lamps are out or all lamps are too dim for the conditions, check the battery or generator for charge. If the charge is satisfactory, but the lamps are still too dim for the conditions, check the photocell is not being blocked from reading the true ambient light level.

B.5.6 Maintenance

Field service

Operators should familiarise themselves with the manufacturer's instructions. A high level of field service is essential to maintain the sign in good condition to provide an effective warning to motorists at all times.

Field service usually covers:

- Simple fault diagnosis and associated replacement of the faulty modules;
- Routine maintenance and servicing including replacement of lamps and other consumable components; and;
- Keeping the equipment clean and tidy.

These requirements are detailed in the manufacturer's instructions.

Major repairs

If major faults develop, the sign should be taken out of service and alternative traffic control arrangements made. The sign should not be returned to service until the faults have been rectified.

Administration procedures

See local office procedures in respect of administration, costing and maintenance of flashing arrow signs as an item of small plant.

B.5.7 Signage

Instructions on the use of illuminated flashing arrows are given in [Section 6.9.2 Illuminated flashing arrow signs](#).

The following three sign size designations are used:

- Type A – 1260 mm by 650 mm for roof mounting on light vehicles;
- Type B – 1500 mm by 770 mm for cab mounting on trucks; and
- Type C – 2400 mm by 1200 mm for trailer mounting and using its own power source or cab mounting on a truck.

High intensity flashing lamps may be used in conjunction with this sign provided that the lamps are either appropriately shielded or laterally or vertically displaced from the edge of the sign to avoid visually corrupting the arrow shape or its directional effect.

Requirements for the flashing of different patterns of the lights are:

- When traffic is expected to pass the sign on a particular side and can do so in safety, i.e. it is not required to seek a gap in oncoming traffic, the bar of the arrow and the barb directing traffic to that side must be flashed; and
- When the sign is used to give a general warning of works activity ahead, including dynamic works, but either the sign is located clear of the traffic path or the display of an arrow would not be appropriate for some other reason, the four corner lights at the extremities of the barbs must be flashed with diagonal pairs flashing alternately.

Additional instructions on the associated signs and use of illuminated flashing arrows are given in [Section 6 Signs and devices](#).

B.6 Usage procedure: temporary portable rumble strips

This Section supplements the requirements outlined in [Section 6.11 Temporary portable rumble strips \(TPRS\)](#) and details the operational requirements for temporary portable rumble strips (TPRS) in NSW. At the time of publication, there is no Transport specification for TPRS.

B.6.1 Installation and operation

Placement, installation and removal

To protect oncoming traffic from the risk of swerving behaviour TPRS must be placed in one of the following methods:

1. Where traffic is reduced to and controlled within a single lane; or
2. On a two-lane two-way road where a temporary median kerb with delineators is installed on the centreline for a distance of D prior to the first strip in each array.

Where placed on a two-lane two-way road, where a temporary median kerb with delineators, the median kerb delineators must be installed:

- Commencing immediately adjacent to the first TPRS;
- Extend a minimum distance of D prior to the TPRS; and
- Have spacing of not greater than 18 m between the centres of each section of kerb.

Example work site layouts for shuttle flow and merge lane closure scenarios using a TPRS are provided in [D.4.10 Static: Use of temporary portable rumble strips \(shuttle flow\)](#) and [D.4.11 Static: Use of temporary portable rumble strips \(merge lane closure\)](#).

Additional examples of work site layouts for TPRS on approach to work area are provided in [Figure 6-34](#) and [Figure 6-35](#).

When installed, TPRS must:

- Be located away from and not adjacent to, workers on foot;
- Be located at least four seconds of travel time after a driver decision point, driver action point or another TPRS array;
- Be located at least 30 m away from the work area;
- Be visible to an approaching road user for at least a distance of D metres; and
- Be installed in accordance with the manufacturers operation manual and guidelines, however, in case of a discrepancy, the requirements of this Technical Manual prevail.

When located on a designated cycle path, a minimum 1 m shoulder adjacent to the TPRS must be provided. If this is not possible, other safe alternative provisions must be made.

If the length of the work area is greater than 200 m, or there is an interrupted line of sight between each end of the work area, then intermediate TPRS arrays and their associated signing should be installed at intervals of not less than 100 m.

When installing TPRS, appropriate manual handling techniques should be used in the set-up, pack up and transportation of the device.

When the work site is unattended or a speed limit of greater than 65 km/h is reinstated, TPRS must be removed and associated signs (T5-210, T5-211) removed or covered up.

Operation

During the operation of TPRS:

- Regular monitoring and review of the TPRS must be undertaken to monitor movement and driver behaviour;
- If adverse driver behaviour is observed, such as heavy braking or swerving on approach or around the TPRS, additional controls such as increasing the TPRS sign's advance warning distance should be implemented;
- In the event of swerving behaviour, the installation of additional control devices such as cones, temporary kerbing or bollards, should be implemented in consideration of OSOM vehicles; and
- If adverse driver behaviour continues to be observed, TPRS must be replaced with an alternative form of traffic control and recorded.

Low visibility and night time operation

When TPRS are operating in low visibility environments such as fog, low light or under night conditions, the area in which each array are being used must be sufficiently lit to ensure the visibility of the TPRS to road users.

Movement

TPRS are not secured to the road surface and as such, movement of the devices is possible. Factors influencing the degree of movement include:

- Type of road surface;
- Grade;
- Traffic composition; and
- Volume.

Regular inspections for TPRS movement must be carried out during operation.

Inspections should be completed more frequently when:

- Using TPRS at a work area for the first time;
- Used on grades;
- Used under high volumes of traffic; or
- Used under a higher proportion of heavy vehicles.

Maximum movement tolerances for TPRS are provided below in [Figure B-9](#) to [Figure B-11](#), and if movement exceeds these tolerances, or if the TPRS becomes disconnected, it must be rectified immediately.

To assist in identifying the extent of movement, a method of determining placement should be employed, such as using a chalk mark or aligning with a landmark to indicate where the TPRS were placed.

The frequency of remediation of displaced or disconnected strips must be monitored and risk-assessed. If it is determined that the frequency and risk of remediation exceeds the benefit gained by the use of the TPRS, the TPRS must be removed.

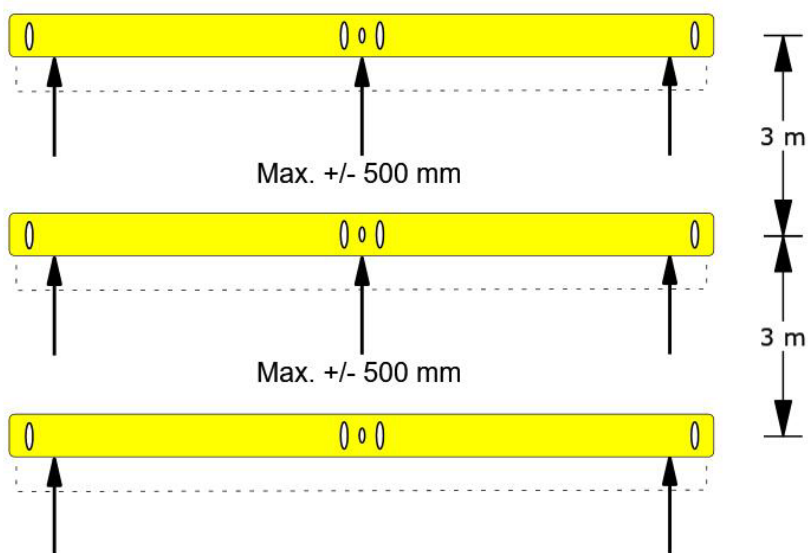


Figure B-9. Maximum movement tolerance for TPRS - Longitudinal movement

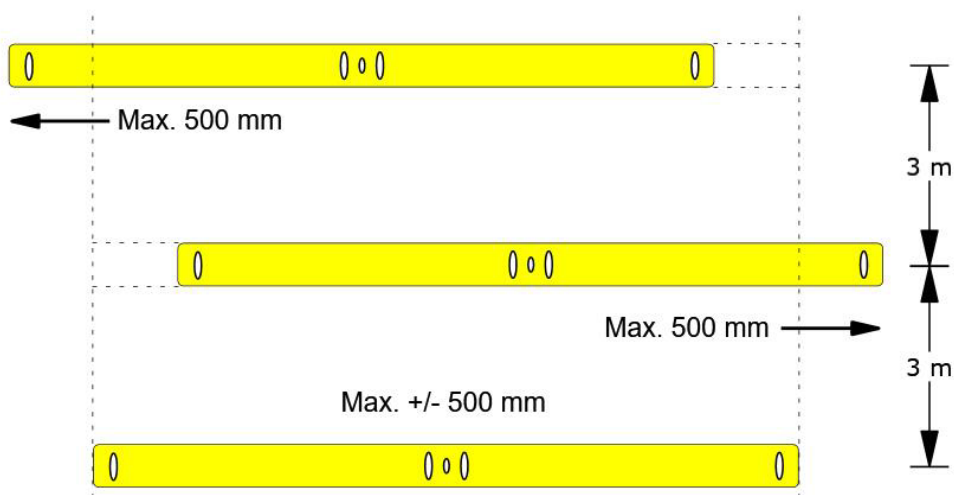


Figure B-10. Maximum movement tolerance for TPRS - Lateral movement, sideways

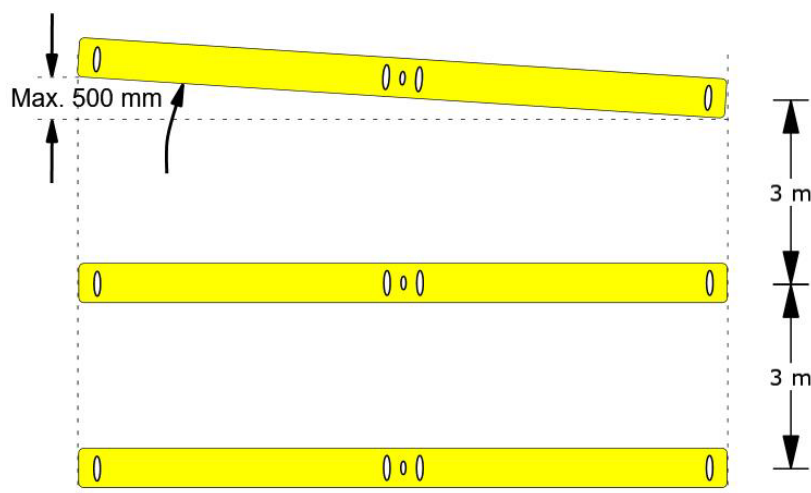


Figure B-11. Maximum movement tolerance for TPRS - Skewing / rotation

B.6.2 Signage

Warning signs must be installed to alert drivers to the use of TPRS.

TPRS must be used with the following signage:

- A Speed hump ahead symbolic (*Figure B-12*) warning sign installed at distance D before the first rumble strip. The distance may be increased to 1.5D – 2D when there is poor compliance or adverse driver behaviours, such as excessive speed or harsh braking; and
- A Speed hump symbolic (*Figure B-13*) warning sign placed next to the first strip at a TPRS array.

When using TPRS with a temporary median kerb and delineators the following additional signage must be installed:

- A temporary collapsible chevron delineator (T5-7) on the first temporary median on approach to the TPRS (see *Figure B.14*); and
- A NO OVERTAKING sign (W8-240n) must be installed at distance 1.5D before the first rumble strip in advance of the temporary median (see *Figure B.15*).



Figure B-12. T5-210 Speed hump ahead sign

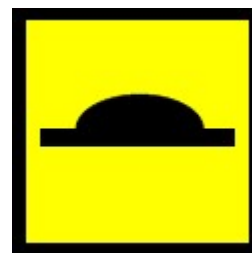


Figure B-13. T5-211 Speed hump sign



Figure B-14. T5-7 Chevron delineator



Figure B-15. W8-240n No Overtaking

When the work site is unattended or a speed limit of greater than 60 km/h is reinstated, TPRS must be removed and associated signs, (*Figure B-13*) and (*Figure B-12*) removed or covered up.

A Variable Message Sign (VMS) may be used to further alert drivers of TPRS if adverse driver behaviour is identified. The addition of VMS should be considered as part of the development of the TMP and included by the designer as part of the TGS.